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## Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

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**Supplementary Materials**

**FAHR-Face (Foundation Model) Training Dataset**

The foundation model was trained on the WebFace42M[1] dataset, a cleaned subset of the WebFace260M dataset[1]. WebFace42M[1] consists of over 40 million high-quality facial images spanning 2 million unique identities, ensuring a diverse and comprehensive training set for machine learning tasks involving faces.

It contains individuals from over 200 distinct countries/regions and more than 500 professions, providing broad diversity in nationality and background. Analysis of the cleaned WebFace42M[1] shows it spans an extensive range of poses (yaw angles from -90 to 90 degrees) and includes most major races worldwide, such as Caucasian, African, East Asian, South Asian, and more[1]. The WebFace42M[1] dataset covers a wide range of ages, with dates of birth going back to 1846, ensuring substantial age diversity in the face images.

The dataset's diversity extends to image quality and lighting conditions. The images were collected from various sources across different time periods and scenarios (both controlled and unconstrained), so they represent a wide spectrum of photo qualities and lighting situations. This variety in image characteristics helps in training models that can generalize well to real-world applications where lighting and image quality may vary significantly.

The foundation model was trained on exactly 40,615,808 face images from approximately 2 million individuals with a size of 112x112 pixels. Besides the processing steps taken during training, no further preprocessing was performed on the images from WebFace42M[1]. The cleaning process used to create WebFace42M[1] from the larger WebFace260M[1] dataset likely removed many low-quality images, ensuring a baseline level of image quality while maintaining diversity in lighting and capture conditions.

**FAHR-Face (Foundation Model) Architecture and Training**

The foundation model is based on the Vision Transformer (ViT) architecture[2], specifically the base-sized model pre-trained using the Masked Autoencoder method. The "facebook/vit-mae-base[3]" model was released in the Facebook AI repository.

The ViT architecture is a transformer encoder model that processes images as a sequence of fixed-size patches. In this implementation, a patch size of 16x16 pixels is used. During pre-training with the Masked Autoencoder, a high portion (75%) of the image patches are randomly masked. The encoder encodes the visible patches, and a learnable mask token is added at the positions of the masked patches. The decoder then reconstructs the raw pixel values for the masked positions based on the encoded visible patches and mask tokens.

Several adjustments were made to adapt the "facebook/vit-mae-base[3]" model for this use case. First, the input images were resized to 112x112 pixels to match the resolution of the face photograph dataset, which altered the number of patches compared to the original 224x224 configuration. The positional embeddings for both the encoder and decoder were interpolated to handle the new patch arrangement while keeping the patch size at 16x16 pixels. The model was then fine-tuned on the WebFace42M[1] dataset using the AdamW optimizer with a learning rate of 1.5e-5, weight decay of 0.05, beta values of (0.9, 0.9999), and a batch size of 1024 over 23 epochs.

The fine-tuning process leverages transfer learning from the pre-trained weights of the "facebook/vit-mae-base[3]" model, allowing our foundation model to learn rich representations specific to facial analysis tasks. The model captures meaningful facial features and structures by utilizing the Masked Autoencoder objective during pre-training, providing a strong basis for subsequent tasks. Training, tuning, and testing were done on a workstation equipped with an AMD Ryzen Threadripper PRO 5975WX CPU, NVIDIA RTX A6000 GPU, and 252 GB RAM.

**FAHR-Face (Foundation Model) Evaluation**

The reconstruction ability of the trained foundation model is demonstrated in **Extended Data Figure 5**, which shows that the model can effectively capture and reconstruct critical facial features, structures, and details across a diverse range of ages, and races/ethnicities. Synthetic facial photographs (n=22) were generated using Midjourney. The foundation model successfully reconstructed 75% of randomly masked regions in each photograph. Notably, the model employs an asymmetric encoder-decoder architecture with a purposely small encoder and larger decoder; this design ensures that the encoder learns compact and meaningful representations, with reconstruction serving as a training mechanism to guide representation learning rather than being the primary objective.

**FAHR-FaceAge (Age Estimation Model) Datasets**

The age estimation model combines eleven diverse facial age datasets to ensure a comprehensive and balanced training and testing set. The selected datasets include:

**IMDB-WIKI[4]:** The IMDB-WIKI dataset is a collection of 523,051 images annotated with age labels calculated based on birth year and photo date, collected from 100,000 celebrity profiles on IMDb and Wikipedia.

**KANFace**[5]**:** The KANFace dataset contains 41,036 images of 1,045 subjects captured in unconstrained, real-world conditions, with manual annotations for age labels.

**FG-NET**[6]: The FG-NET dataset comprises 1002 face images from 82 individuals captured at different ages. Age label provided directly by the subjects.

**CACD**[7]**:** The Cross-Age Celebrity Dataset consists of 163,446 face images of 2,000 celebrities collected from the Internet. The dataset was compiled using a combination of the celebrity's name and year (from 2004 to 2013) as search keywords, allowing the images to be labeled with age label for each celebrity based on the year the photo was taken and their known birth year.

**AFAD**[8]**:** The Asian Face Age Dataset is a large-scale face image dataset. It consists of 165,501 facial images captured from Asian individuals and corresponding age (calculated with photo upload date and date of birth) and sex label. The AFAD was collected to address the lack of an extensive dataset for evaluating age estimation performance on Asian faces.

**MegaAge**[9]**:** The MegaAge dataset consists of 41,941 face images collected from the Internet with age labels. The dataset was compiled using an approach that transfers reliable age annotations from the FG-NET database to unlabeled images via human-in-the-loop comparisons. Age labels derived from posteriors of age comparisons.

**MORPH**[10]**:** The MORPH dataset contains 55,134 images with age labels of 13,617 individuals. The dataset was collected from public records including birth date and date of photo captured.

**LAGENDA**[11]**:** The Layer Age and Gender Dataset consists of 67,159 images with 83,144 annotated face images sourced from the Open Images Dataset, covering various real-world scenes and domains with minimal celebrity data. Age labels from weighted crowd-vote consensus.

**UTK**[12]**:** The UTKFace dataset is a face image dataset that spans a wide age range. It consists of 24,108 face images with annotations for age. A deep learning model inferred age, followed by human quality assurance. The dataset includes unconstrained faces with pose, expression, illumination, occlusion, and resolution variations.

**AgeDB**[13]**:** The AgeDB dataset is manually collected, comprising 16,488 in-the-wild images of well-known individuals like actors, scientists, and politicians, each annotated with age a total of 568 distinct subjects. Age labels were manually collected and verified to the exact year from source captions.

**APPA-REAL**[14]**:** The APPA-REAL dataset was created by combining community inputs through a web application and data from the AgeGuess platform, resulting in 7,591 images with age labels. The dataset consists primarily of photos contributed by the public rather than celebrities. Age labels came from participant self-reports or verified public metadata.

The photos from IMDB-WIKI[4], FG-NET[6], CACD[7], MegaAge[9], MORPH[10], UTK[12], AgeDB[13], APPA-REAL[14] were cropped and aligned with RetinaFace[15]; only photos where exactly one face was detected were used. For LAGENDA[11], KANFace[5], and AFAD[8], the already cropped images provided in these datasets were used.

The dataset was divided into three categories: Pre-training, which includes IMDB-WIKI[4], KANFace[5], FG-NET[6], CACD[7], AFAD[8], MegaAge[9], MORPH[10], LAGENDA[11]; Fine-tuning, comprised of UTK[12], AgeDB[13] (chosen specifically for their more balanced age distribution and representation of older subjects); Evaluation, focused on APPA-REAL[14] (chosen specifically for its more balanced age distribution, representation of older subjects, and primary contribution from the general public, making overlap with other datasets unlikely). **Extended Data Figure 6** illustrates the dataset filtering and preprocessing pipeline applied to each dataset before combining them into the final training, fine-tuning, and evaluation sets. The pipeline begins with the raw datasets, each containing a unique set of face photos with corresponding age labels. After the filtering and preprocessing steps, the age distribution of each dataset is analyzed and visualized in **Extended Data Figure 6.**

The multi-dataset approach in the training data promotes high generalization, allowing the model to perform well on unseen faces and adapt to various real-world scenarios.

**FAHR-FaceAge Model Adjustments and Training**

The FAHR-FaceAge was built upon our foundation model, a Vision Transformer (ViT) pre-trained using the Masked Autoencoder method. The model architecture was modified to include a regression head for age estimation, consisting of a single linear layer that maps the hidden state of the first token (CLS token) to a scalar age value.

Data augmentation techniques were applied during training to enhance the model's robustness and generalization ability. The augmentation steps included Random Resized Crop with size (112, 112) and scale (0.8, 1.0), Random Horizontal Flip with a probability of 0.5, Color Jitter with brightness 0.4, contrast 0.4, saturation 0.4, hue 0.1, and probability of 0.8, Random Grayscale with probability 0.2, Random Rotation with 10 degrees, Gaussian Blur with kernel size (5, 5), sigma (0.1, 2.0), and probability of 0.5, Random Erasing with probability of 0.5, scale (0.02, 0.33), ratio (0.3, 3.3), and value 0, and Normalization with mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].

The training process was divided into two stages: pre-training and fine-tuning. The model was trained on a combination of eight diverse facial age datasets described above for pre-training. These datasets collectively provided various face images with corresponding age labels. However, the age distribution in these datasets was imbalanced, with certain age groups (e.g., elderly individuals) underrepresented.

An oversampling strategy was applied during pre-training and fine-tuning to correct the imbalance and ensure the model learned from a more evenly distributed age range. The oversampling factors for pre-training were determined based on the age range, with more aggressive oversampling applied to the underrepresented age groups. Images of individuals aged 0-50 years were not oversampled (factor = 1), while images of individuals in the 50-55 age range were oversampled by a factor of 2. The oversampling factor gradually increased for older age ranges, with the 95-116 age range having the highest oversampling factor of 20. This oversampling ensured that the model was exposed to sufficient samples from the underrepresented and rare age groups during pre-training.

During pre-training, the model was optimized using the AdamW optimizer with a base learning rate of 1e-5, weight decay of 0.05, beta values of (0.9, 0.999), and a layer-wise learning rate decay of 0.75. The model was trained for 30 epochs with a batch size of 64. The mean absolute error (MAE) was used as the loss function during pre-training.

The model was further fine-tuned on the UTK[12] and AgeDB[13] datasets. A different oversampling strategy was employed during fine-tuning to ensure a more even distribution of ages. The fine-tuning dataset was divided into age bins of 5 years, and each bin was balanced to a specified size of 200 samples. If a bin contained more than 200 samples, a random subset of 200 samples was selected. A bin with fewer than 200 samples was oversampled by randomly duplicating samples until the desired size was reached. This binning and balancing approach guaranteed that each age group had equal representation during fine-tuning. Both oversampling strategies, pre-training and fine-tuning, are shown in **Extended Data Figure 6.**

The fine-tuning stage utilized the AdamW optimizer with a reduced base learning rate of 1e-6, weight decay of 0.05, beta values of (0.9, 0.999), and a layer-wise learning rate decay of 0.65. The model was fine-tuned for 10 epochs with a batch size of 8, using the MAE loss as the optimization objective.

By employing these oversampling strategies, hyperparameter settings, and the custom learning rate schedule, FAHR-FaceAge was able to learn effectively from the imbalanced pre-training datasets and achieved improved performance on the fine-tuning datasets.

**FAHR-FaceAge Evaluation**

FAHR-FaceAge was evaluated on the public APPA-REAL[14] and Harvard RT patient datasets. Our model's age estimation performance in these two distinct datasets (real-world healthy individuals and cancer patients) was compared against several models representing a diverse range of state-of-the-art architectures and have been widely used in computer vision tasks, including VGG16[16], SENet50[17], DenseNet121[18], and four different versions of MobileNet[19] (V2-96, V2-224, V3-small, and V3-large) as implemented by Greco et al.[20]. These architectures were trained either through knowledge distillation (using VGGFace2[21] images with age labels generated by a teacher model to create VMAGE) or with IMDB-Wiki[4] dataset, and evaluated on the same cropped face photographs as FAHR-FaceAge. While FAHR-FaceAge uses an input size of 112×112 pixels for models that support higher resolution inputs (e.g., 224×224), we resized the images accordingly to leverage their full capabilities.

For the APPA-REAL dataset, the MAE comparison showed that FAHR-FaceAge has the lowest MAE at 5.1, followed by SENet50 VMAGE at 5.5, MobileNet V3-large VMAGE at 5.6, MobileNet224 VMAGE at 5.9, MobileNet V3-small VMAGE at 5.9, MobileNet96 VMAGE at 6.1, DenseNet121 VMAGE at 6.4, DenseNet IMDB-WIKI at 7.9, MobileNet V3-large IMDB-WIKI at 8.0, MobileNet V3-small IMDB-WIKI at 8.0, MobileNet224 IMDB-WIKI at 8.2, MobileNet96 IMDB-WIKI at 8.6, SENet50 IMDB-WIKI at 8.8, FaceAge[22] at 12.6, VGG16 IMDB-WIKI at 16.5, and VGG16 VMAGE at 63.7.

For the Harvard RT patient dataset, the MAE comparison showed that FaceAge[22] has the lowest MAE at 7.2, followed by FAHR-FaceAge at 7.5, MobileNet V3-large IMDB-WIKI at 9.4, SENet50 IMDB-WIKI at 10.6, SENet50 VMAGE at 11.3, MobileNet V3-large VMAGE at 11.7, MobileNet224 VMAGE at 11.7, DenseNet IMDB-WIKI at 11.8, MobileNet V3-small IMDB-WIKI at 12.1, MobileNet V3-small VMAGE at 12.2, Mobilenet96 VMAGE at 12.5, DenseNet121 VMAGE at 13.2, MobileNet224 IMDB-WIKI at 13.3, MobileNet96 IMDB-WIKI at 14.5, VGG16 IMDB-WIKI at 27.1, and VGG16 VMAGE at 30.5.

In addition to reporting the MAE across all samples, we evaluated each model using the average bin-wise MAE: for each 5-year age bin (starting at age 0), we calculated the MAE and then averaged these values across all bins. This metric better captures model robustness to age bias and reflects overall accuracy across the lifespan, penalizing models that perform poorly in specific age groups.

For the APPA-REAL dataset, the Average bin-wise MAE comparison showed that FAHR-FaceAge has the lowest value at 5.7, followed by SENet50 VMAGE at 9.7, MobileNet V3-large VMAGE at 10.0, MobileNet224 VMAGE at 10.2, MobileNet V3-small VMAGE at 10.4, MobileNet96 VMAGE at 10.7, DenseNet121 VMAGE at 11.0, MobileNet V3-large IMDB-WIKI at 11.2, MobileNet224 IMDB-WIKI at 11.7, FaceAge[22] at 11.8, DenseNet IMDB-WIKI at 12.1, SENet50

IMDB-WIKI at 12.3, MobileNet V3-small IMDB-WIKI at 12.4, MobileNet96 IMDB-WIKI at 12.7, VGG16 IMDB-WIKI at 21.6, and VGG16 VMAGE at 45.7.

For the Harvard RT patient dataset, the Average bin-wise MAE comparison showed that FAHR-FaceAge has the lowest value at 6.1, followed by SENet50 VMAGE at 10.5, MobileNet V3-large VMAGE at 10.8, MobileNet V3-large IMDB-WIKI at 11.0, MobileNet224 VMAGE at 11.0, FaceAge[22] at 11.3, MobileNet V3-small VMAGE at 11.3, MobileNet96 VMAGE at 11.8, DenseNet121 VMAGE at 11.9, DenseNet IMDB-WIKI at 12.3, MobileNet V3-small IMDB-WIKI at 12.7, SENet50 IMDB-WIKI at 13.0, MobileNet224 IMDB-WIKI at 13.2, MobileNet96 IMDB-WIKI at 14.5, VGG16 IMDB-WIKI at 22.4, and VGG16 VMAGE at 43.8.

The 5 best-performing models for each dataset are visualized in **Extended Data Figure 1**.

**Benchmark on Fine-tuned Models**

To ensure a fair comparison, we fine-tuned each competing architecture using the same training configuration as FAHR-FaceAge. Most training parameters, including batch size, number of epochs, and dataset composition (oversampled UTK and AgeDB), remained identical; only the learning rate was adjusted, and augmentation pipeline was changed. Fine-tuning these pretrained models proved non-trivial, as they had initially learned from highly age-biased datasets, primarily consisting of young adult faces. Expanding the target age range to span the full human lifespan, from young children to elderly adults, increases the complexity of the age-estimation task, inherently leading to higher MAE values compared to narrower age-range benchmarks. Pilot experiments indicated that higher learning rates quickly disrupted the pretrained age-specific features, resulting in poor convergence and generalization. Therefore, we carefully selected a conservative learning rate of 1e-7 to ensure stable training and optimal performance across the entire age spectrum. During training, each image undergoes the following augmentations: random rotation, random affine skew, random crop (by jittering the region of interest), random brightness and contrast adjustment, and random horizontal flip (p=0.5). These augmentations are the default augmentation from Greco et al.[20] and applied to every image in training.

Despite this harmonized fine-tuning process, FAHR-FaceAge consistently outperformed the benchmarked models across both evaluation datasets in terms of accuracy and bias, demonstrating the advantage of starting from a large-scale, self-supervised foundation model.

For the APPA-REAL dataset, the MAE comparison showed that FAHR-FaceAge has the lowest MAE at 5.1, followed by MobileNet224 VMAGE at 6.5, SENet50 VMAGE at 6.9, MobileNet V3 large VMAGE at 7.0, MobileNet96 VMAGE at 7.1, DenseNet IMDB-WIKI at 7.9, MobileNet96 IMDB-WIKI at 8.6, MobileNet224 IMDB-WIKI at 9.0, SENet50 IMDB-WIKI at 9.0, MobileNet V3 large IMDB-WIKI at 9.2, MobileNet V3 small IMDB-WIKI at 9.4, DenseNet121 VMAGE at 10.0, FaceAge[22] at 12.6, VGG16 IMDB-WIKI at 14.8, VGG16 VMAGE at 15.2, and MobileNet V3 small VMAGE at 30.3.

For the Harvard RT patient dataset, the MAE comparison showed that FaceAge[22] has the lowest MAE at 7.2, followed by FAHR-FaceAge at 7.5, SENet50 IMDB-WIKI at 8.7, DenseNet IMDB-WIKI at 10.9, SENet50 VMAGE at 12.6, MobileNet V3 small VMAGE at 13.5, MobileNet V3 large IMDB-WIKI at 14.3, MobileNet V3 large VMAGE at 14.4, DenseNet121 VMAGE at 14.5, MobileNet224 VMAGE at 14.9, MobileNet224 IMDB-WIKI at 15.7, Mobilenet96 VMAGE at 16.7, MobileNet96 IMDB-WIKI at 16.9, MobileNet V3 small IMDB-WIKI at 17.8, VGG16 IMDB-WIKI at 18.1, and VGG16 VMAGE at 18.4.

For the APPA-REAL dataset, the Average bin-wise MAE comparison showed that FAHR-FaceAge has the lowest value at 5.7, followed by SENet50 VMAGE at 10.3, SENet50 IMDB-WIKI at 10.7, FaceAge[22] at 11.8, DenseNet IMDB-WIKI at 12.1, MobileNet224 VMAGE at 12.3, MobileNet V3-large VMAGE at 12.4, DenseNet121 VMAGE at 13.1, MobileNet224 IMDB-WIKI at 13.5, MobileNet96 IMDB-WIKI at 13.7, MobileNet96 VMAGE at 14.0, MobileNet V3-large IMDB-WIKI at 14.8, VGG16 VMAGE at 14.8, MobileNet V3-small IMDB-WIKI at 16.7, VGG16 IMDB-WIKI at 16.8, and MobileNet V3-small VMAGE at 48.4.

For the Harvard RT patient dataset, the Average bin-wise MAE comparison showed that FAHR-FaceAge has the lowest value at 6.1, followed by SENet50 VMAGE at 10.7, FaceAge[22] at 11.3, SENet50 IMDB-WIKI at 11.3, DenseNet IMDB-WIKI at 12.0, MobileNet V3-small VMAGE at 12.3, MobileNet V3-large VMAGE at 12.7, MobileNet224 VMAGE at 12.8, DenseNet121 VMAGE at 13.4, VGG16 VMAGE at 14.2, MobileNet96 VMAGE at 14.4, VGG16 IMDB-WIKI at 14.5, MobileNet V3-large IMDB-WIKI at 15.0, MobileNet224 IMDB-WIKI at 15.3, MobileNet96 IMDB-WIKI at 15.4, and MobileNet V3-small IMDB-WIKI at 17.5.

The 5 best-performing models for each dataset are visualized in **Extended Data Figure 1**.

**FAHR-FaceAge Robustness Evaluation Datasets**

We evaluated our model's robustness across three publicly available datasets that capture different types of variations in facial appearance or facial photograph acquisition:

**HDA Plastic Surgery Database**[23]**:** Contains 638 image pairs (540 female, 98 male) showing faces before and after five common plastic surgery procedures: eyebrow correction, eyelid correction, facelift, facial bones correction, and nose correction. All images were collected from multiple web sources. Each image pair shows the same individual before and after a single type of surgical procedure. 631 pairs were used in the analysis (7 pairs were excluded due to face detection failure or inconsistent file naming).

**Makeup Dataset**[24]: Consists of 26 paired images of female subjects with and without makeup application. The dataset includes both subtle and strong makeup changes. All images were collected under controlled conditions with consistent lighting and frontal pose.

**FEI Face Database**[25]**:** A controlled face pose dataset containing 200 subjects (100 male, 100 female) with 14 different pose angles per subject. From the original 2800 images, we used a subset focusing on three key variations: frontal face comparisons with expression changes (neutral and smiling), illumination variations, and pose variations (22.5° left/right vs. frontal).

All images from the three datasets were pre-processed using RetinaFace for face detection and alignment to ensure consistent facial presentation and image quality.

**FAHR-FaceAge Robustness Evaluation**

FAHR-FaceAge robustness to facial alterations and viewing conditions was evaluated using paired images processed with RetinaFace for face detection and alignment. For each image pair, the mean difference and mean absolute difference in age estimation were calculated by subtracting the first condition from the second. Statistical significance was tested using paired Wilcoxon tests.

For plastic surgery evaluation (HDA database[23]), the results were: eyebrow correction (mean difference=0.29±6.30 years, mean absolute difference=4.62 years, n=126), eyelid correction (mean difference=-0.45±5.85 years, mean absolute difference=4.26 years, n=130), facelift (mean difference=-0.40±5.64 years, mean absolute difference=4.53 years, n=96), facial bones correction (mean difference=1.03±6.16 years, mean absolute difference=4.63 years, n=105), and nose correction (mean difference=1.11±4.21 years, mean absolute difference=3.13 years, n=174). For makeup analysis[24], we found a mean difference=1.09±6.31 years and a mean absolute difference=4.37 years (n=26). The FEI database[25] showed viewing angle differences (left: mean difference=0.25±1.49 years, mean absolute difference=1.12 years; right: mean difference=0.55±1.65 years, mean absolute difference=1.17 years; n=200), expression changes (mean difference=0.79±3.68 years, mean absolute difference=2.19 years, n=200), and illumination variations (mean difference=0.55±3.77 years, mean absolute difference=2.35 years, n=181).

**Random initialization baseline**

To isolate the contribution of foundation-model pre-training, we trained a ViT with random weight initialization. Architecture, optimizer hyper-parameters, data splits, augmentations, batch size, and number of epochs were kept identical to FAHR-FaceAge; the sole change was disabling layer-wise learning-rate decay.

The baseline ViT trained from random initialization reached 6.92 years MAE (ME 1.29 years) on APPA-Real and 9.11 years MAE (ME 0.25 years) on the Harvard RT patient dataset.

**FAHR-FaceSurvival (Risk Estimation Model) Dataset Considerations**

The Harvard RT patient dataset (described in further data below) was randomly split into two parts: 90% for training and 10% for testing. During the training process, 5% of the training data was used for validation. The test dataset is not used for validation, and the model never sees this test dataset during training or validation.

**FAHR-FaceSurvival (Risk Estimation Model) Model Integration and Architecture**

A custom dataset loader was implemented to handle the dataset loading and preprocessing. It read the survival data from a CSV file and loaded the corresponding face photos from a specified folder. The dataset returned each sample's face photo, survival time, and outcome (death or censored) status.

Data augmentation techniques were applied to the training set to improve the model's robustness and generalization ability. The validation set underwent resizing and normalization. The specific augmentations and preprocessing steps included Random Resized Crop with size (112, 112) and scale (0.9, 1.0), Random Horizontal Flip with a probability of 0.5, Color Jitter with brightness 0.2, contrast 0.2, saturation 0.2, hue 0.1, and a probability of 0.8, Random Rotation with 5 degrees, and Normalization with mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].

The FAHR-Face was initialized with the pre-trained weights. The survival head was a linear layer that mapped the hidden representation of the first token (CLS token) to a scalar FAHR-FaceSurvival. The model architecture allowed for fine-tuning the pre-trained weights while learning the task-specific risk estimation.

The training process utilized a custom ranking loss function designed to optimize the ranking of the FAHR-FaceSurvival based on survival times. To achieve this, the loss function first identified pairs of patients within each training batch, where each pair consisted of one patient who experienced the event of interest (death), thus representing a complete observation, and another patient whose survival time was observed to be longer. All possible pairs in each batch were used. The loss function encouraged the model to assign a higher FAHR-FaceSurvival to patients with shorter survival times compared to those with longer survival times. The loss function also incorporated a smoothed regularization term to encourage smooth FAHR-FaceSurvival predictions. The smoothed regularization term in the ranking loss function promoted the model to produce FAHR-FaceSurvival predictions that were smooth and consistent across similar patients by penalizing large differences between the FAHR-FaceSurvival of adjacent patients in the sorted order, promoting a more stable and coherent risk prediction.

During training, the model was optimized using the AdamW optimizer with a base learning rate of 1e-5, weight decay of 0.05, beta values of (0.9, 0.999), layer-wise learning rate decay of 0.75, batch size of 32, and smoothed regularization strength of 1e-4. The model was trained for 10 epochs, and checkpoints were saved after each epoch. The model's performance was evaluated using a custom concordance index (C-index) function, which measured the model's ability to rank the FAHR-FaceSurvival correctly. The C-index was computed separately for each epoch's training and validation sets. The training and validation losses were also monitored and reported.

**FAHR-FaceSurvival (Risk Estimation Model) Evaluation**

FAHR-FaceSurvival was evaluated on two datasets: the Harvard RT patient test dataset and the Maastro RT patient dataset. FAHR-FaceSurvival was min-max scaled to a range between 0 and 1 for each dataset to ensure consistency and facilitate comparative analysis.

**Study Population Harvard RT Patient Dataset**

The Harvard RT patient dataset consists of 38,183 patients treated with radiation therapy for cancer or benign conditions between 2008 and 2023 at six Harvard-affiliated radiation oncology clinical sites: Brigham and Women's Hospital, Dana-Farber Cancer Institute, Mansfield, Milford, South Shore Hospital/Dana-Farber Cancer Institute, and South Shore Hospital/Dana-Farber Cancer Institute Libbey Park. All facial photographs were obtained before the start of radiotherapy. Diagnosis associated with the radiation therapy course for each patient was obtained based on the international classification of diseases ICD-9 or ICD-10 codes. ICD codes were categorized into diagnosis and cancer site groups used in the study analysis, as detailed in **Extended Data Table 4**. Following the classification of diagnosis based on the ICD codes, available case descriptions summarizing the diagnosis, including stage and radiation plan, were used to categorize cases further. For example, if a patient had an ICD-10 code of C61 indicating prostate cancer, but the case description indicated "metastatic" prostate cancer, then the diagnosis was re-classified as metastatic disease. For oropharyngeal cancers, if the description indicated "HPV+ (human papillomavirus)" type, then the diagnosis was further specified as HPV+ oropharyngeal cancer. Benign diseases included keloids, heterotopic ossification, non-malignant brain tumors (pituitary adenomas, schwannomas, meningiomas), brain arteriovenous malformation, and cardiovascular disease (ventricular tachycardia, coronary artery disease).

From the initial 46,770 patients, 927 photos were excluded due to poor quality or substantial portions of the face being hidden (e.g., face masks, sunglasses), resulting in 45,843 remaining patients. The photos of these 45,843 patients were processed using the RetinaFace[15] face detection and alignment algorithm, retaining only those where exactly one face was detected, resulting in a dataset of 44,182 patients (1,661 patients excluded). Patients with missing follow-up information or cancer diagnosis were further excluded (5,999 patients), leading to a final cohort of 38,183 patients for analysis. Detailed characteristics of the Harvard RT patient dataset are provided in **Extended Data Table 1**.

**External Validation – Maastro RT patient Dataset**

The Maastro RT patient dataset consists of 4,892 patients with cancer diagnoses, prospectively collected and included in the Maastro Biobank (Maastricht, The Netherlands). All facial photographs were obtained before the start of radiotherapy. These patients were treated with radiation therapy with curative or palliative intent between 2006 and 2019. The predominant primary malignancies were breast, colorectal, prostate, lung, and head and neck cancer. From an initial dataset of 6,835 patients, we excluded patients with missing face photographs, duplicate entries, and lack of follow-up information. A manual image quality assessment was performed on the remaining entries; patients were excluded due to poor image quality, leaving us with 5,498 patients. Patients with metastatic disease, those treated with palliative intent, or those diagnosed with ductal carcinoma in situ of the breast (DCIS) were subsequently removed from the dataset. This step reduced the cohort to 4,906 patients.

In the final analysis, 4,892 patients were included. This final selection was based on the use of a face detection model, RetinaFace[15] that only selected photos with precisely one person, ensuring consistency in FAHR-FaceAge and FAHR-FaceSurvival predictions. The dataset includes information on patient demographics, cancer site, treatment details, and survival outcomes. Detailed characteristics of the Maastro patient dataset are provided in **Extended Data Table 1**.

**Random initialization baseline**

To isolate the contribution of foundation-model pre-training, we trained a ViT with random weight initialization. Architecture, optimizer hyper-parameters, data splits, augmentations, batch size, and number of epochs were kept identical to FAHR-FaceSurvival; the sole change was disabling layer-wise learning-rate decay.

**Interpretability, Independence, and Biological Basis of the Models – 2D and 3D Attention Maps**

Attention maps from the trained FAHR-FaceAge and FAHR-FaceSurvival models were computed to identify which facial regions contributed most to model predictions. For 112×112 pixel inputs, self-attention operates on a grid of 7×7 image patches, so raw attention provides a low-resolution signal that was refined through interpolation and averaging across the Harvard RT patient test dataset.

Images from the Harvard RT patient dataset were resized to 112×112 pixels, normalized, and passed through both models. From the final self-attention layer, we extracted the attention from the class token to all image patches and averaged across attention heads to obtain one 7×7 attention map per image. These maps were resampled to the full image resolution using bilinear interpolation for visualization. Average two-dimensional attention maps for each model were then obtained by averaging the per image attention maps across all processed images.

To obtain a spatially accurate view on the face, we mapped attention onto a three-dimensional facial mesh. For each image, facial landmarks were detected using MediaPipe Face Mesh, yielding 468 landmarks in image coordinates that were used to align a canonical three-dimensional face model. Of the 3,820 photographs, 3,815 were successfully processed and five were excluded because no facial landmarks could be detected. The mesh was refined by one round of triangle subdivision (each triangle split into four), with landmark positions for the new vertices obtained by linear interpolation. The aligned mesh was then projected into the image plane, and for every triangle we defined a polygon on the attention map, created a mask of its area, and computed the mean attention value inside that area to obtain one attention score per triangle and per image. Triangle scores were averaged across all images, mapped to colors, and assigned to the corresponding triangles, and the resulting colored three dimensional face meshes were exported in OBJ format with vertex positions and RGB values.

**Statistical Analysis**

Descriptive statistics were used to summarize baseline patient characteristics in the two patient cohorts. For survival analysis, the primary endpoint was overall survival. Survival time was measured from the date the facial photograph was obtained to the date of death or the last documented follow up, with patients still alive censored at their last follow up date. The proportional hazards assumption was tested for all Cox models. Cause of death was not available, so all analyses used overall survival (all-cause mortality) as the endpoint. Univariate Cox analysis was performed to assess the effect of FAHR-FaceAge or FAHR-FaceSurvival variables as well as clinical covariables including age, sex, race/ethnic group, diagnosis, treatment course intent, year treated, and radiation technique on mortality risk. A multivariate Cox model was generated incorporating FAHR-FaceAge or FAHR-FaceSurvival variables and all covariables with P-value <0.05 on univariate analyses. FAHR-FaceAge parameters assessed were FAHR-FaceAge as a continuous variable (per decade), and the deviation between FAHR-FaceAge and chronological age (FAD) as a continuous variable (per decade), FAD ≥5 vs <5, and FAD ≤-5 vs >-5. FAHR-FaceSurvival parameters assessed were FAHR-FaceSurvival as a continuous variable (per 0.1 from zero to one), FAHR-FaceSurvival as quartiles (<0.25 vs. 0.25-0.49, 0.50-0.74 vs. ≥0.75), and FAHR-FaceSurvival as a binary split (<0.5 vs. ≥0.5).

Detailed results of the univariable and multivariate Cox regression analyses for FAHR-FaceAge and FAHR-FaceSurvival within various age, racial, and clinical subgroups in the Harvard RT patient dataset are provided in **Extended Data Table 2** and **Extended Data Table 3**, respectively, and for the Maastro RT patient dataset in **Figure 4** respectively.

Harrell's concordance index (C-index) was utilized to evaluate the predictive performance of the FAHR-FaceAge model in **Figure 2c** for the Harvard RT patient dataset and of the FAHR-FaceSurvival in **Figure 3g** for the Harvard RT patient test dataset. Subgroup analyses evaluated the model's performance across different demographic and clinical groups, including age groups, racial and ethnic categories, and cancer diagnosis.

**Area Under the Curve (AUC) Analysis**

The predictive performance of the FAHR-FaceAge was benchmarked against a prior deep learning-based facial age estimation model FaceAge[22], for the clinical application of survival prediction by using time-dependent receiver operating characteristic (ROC) curves and the area under the curve (AUC) metric in the Maastro RT patient dataset and Harvard RT patient dataset.

Data were extracted from Maastro RT patient dataset and Harvard RT patient dataset, focusing on the subset of patients older than 60 years because the prior FaceAge[22] model was optimized for individuals over 60.

Time points of interest were set at 3 months, 6 months, 1 year, and 2 years after baseline. Time-dependent ROC curves were constructed using the Kaplan-Meier estimator method for each time point. The predictors evaluated

included chronological age, FaceAge (age estimated by the FaceAge model[22]), and FAHR-FaceAge. The AUC was calculated for each predictor and time point to assess the model's discriminative ability at that specific time.

**Analysis of Independence of Imaging Biomarkers from the Two Models**

To demonstrate the independence of FAHR-FaceAge and FAHR-FaceSurvival despite their shared foundation model, we analyzed their correlation in both development and external validation cohorts. In the Harvard RT patient test dataset, FAD and FAHR-FaceSurvival showed a weak positive correlation (R=0.218; **Extended Data Figure 4d**). This independence was further validated in the Maastro RT patient dataset, where FAD and FAHR-FaceSurvival maintained a low correlation (R = 0.160; **Figure 4e**), suggesting these measures capture distinct aspects of patient health even in an independent cohort.

The feature space independence was quantified by computing the cosine similarity between the final hidden state representations (768-dimensional class token embeddings) of both models for each patient image. Cosine similarity measures the cosine of the angle between two vectors, ranging from -1 (completely opposite) to 1 (exactly aligned), with 0 indicating orthogonality. The median similarity of -0.018 across all test patients indicates near-orthogonal feature spaces, suggesting the models learned to extract distinct and complementary features despite sharing the same foundation model architecture (**Extended Data Figure 4e)**.

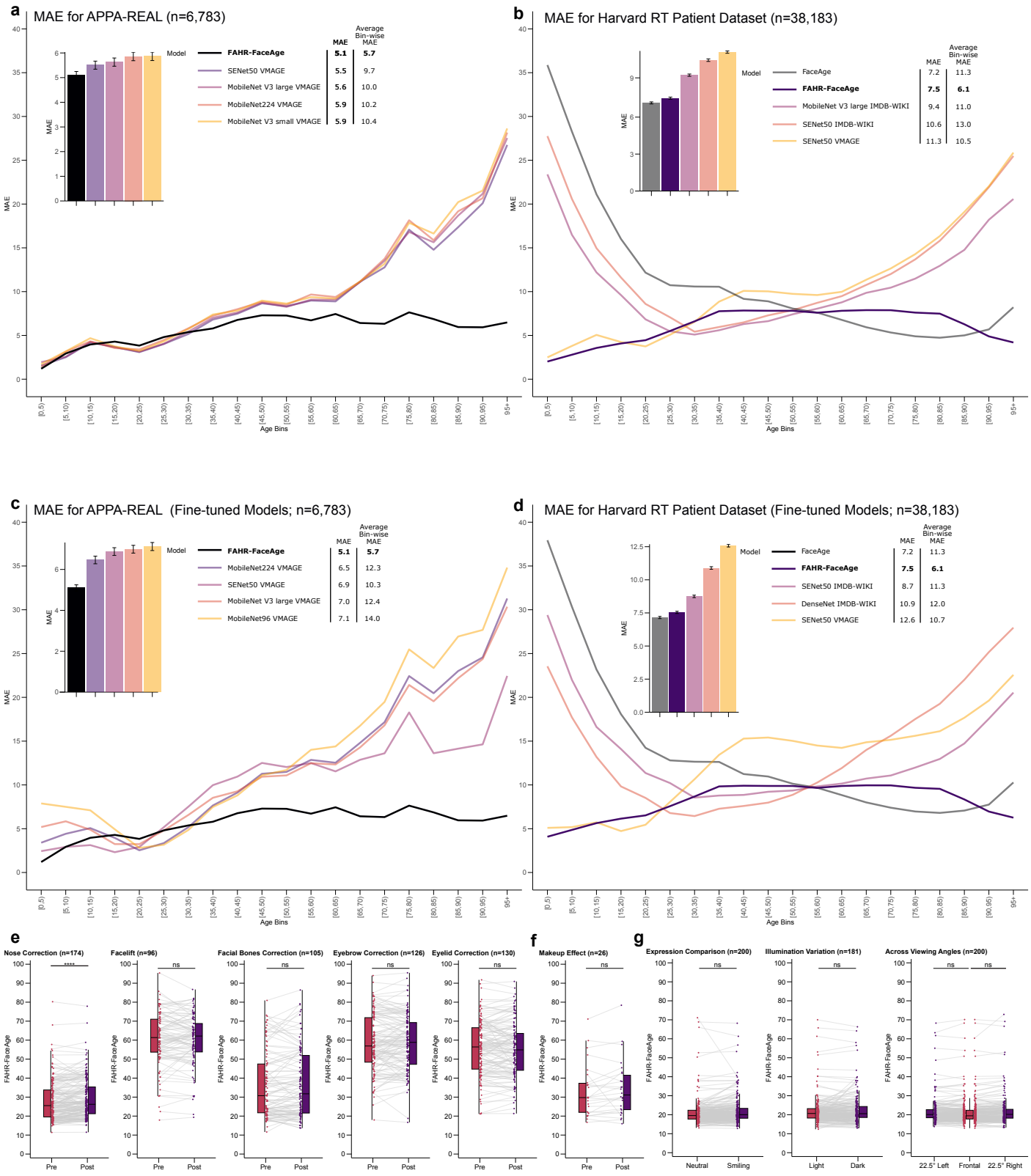
To assess the incremental prognostic value of our imaging biomarkers, we conducted a model comparison using AIC. Three Cox proportional hazards models were fitted: one including FAD as a continuous variable (per decade), one including FAHR-FaceSurvival as a continuous variable (per 0.1 unit increase), and a combined model incorporating both biomarkers, with adjustments for age (in decades), sex, and cancer site. AIC was calculated as AIC = -2 log(L) + 2k, where L is the likelihood of the model and k is the number of estimated parameters; lower AIC values denote a better trade-off between model fit and complexity.

All statistical analyses were performed using R Statistical Software v4.3.1. Statistical significance was defined as a two-sided P-value <0.05

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**Extended Data Figure 1: State-of-the-Art Comparison for Age Estimation Models and Robustness Test**

Each panel presents the performance of the top five age estimation models from face photographs, demonstrating the FAHR-FaceAge model's performance in both mean absolute error (MAE) and average bin-wise MAE across different datasets. Notably, for the Harvard RT patient dataset, FaceAge achieves a lower MAE due to its optimization for older patients (≥60 years), which aligns with the age distribution of the cancer cohort. However, FaceAge does not rank among the top five models for the APPA-REAL dataset, which is attributed to this dataset's younger age distribution. MAE is reported in years. Error bars represent 99 % confidence intervals.

**a,** MAE for the APPA-REAL dataset, shown as a bar plot. The line plot displays performance for each 5-year age bin for the top five models. The models included are FAHR-FaceAge, SENet50 VMAGE, MobileNet V3-large VMAGE, MobileNet224 VMAGE and MobileNet V3-small VMAGE.

**b,** MAE for the Harvard RT patient dataset, shown as a bar plot. The line plot displays performance for each 5-year age bin for the top five models. The models included are FaceAge, FAHR-FaceAge, MobileNet V3-large IMDB-WIKI, SENet50 IMDB-WIKI and SENet50 VMAGE.

**c,** MAE for the APPA-REAL dataset after fine-tuning on age-balanced dataset, shown as a bar plot. The line plot displays the performance for each 5-year age bin for the top five models. The models included are FAHR-FaceAge, MobileNet224 VMAGE, SENet50 VMAGE, MobileNet V3-large VMAGE and MobileNet96 VMAGE.

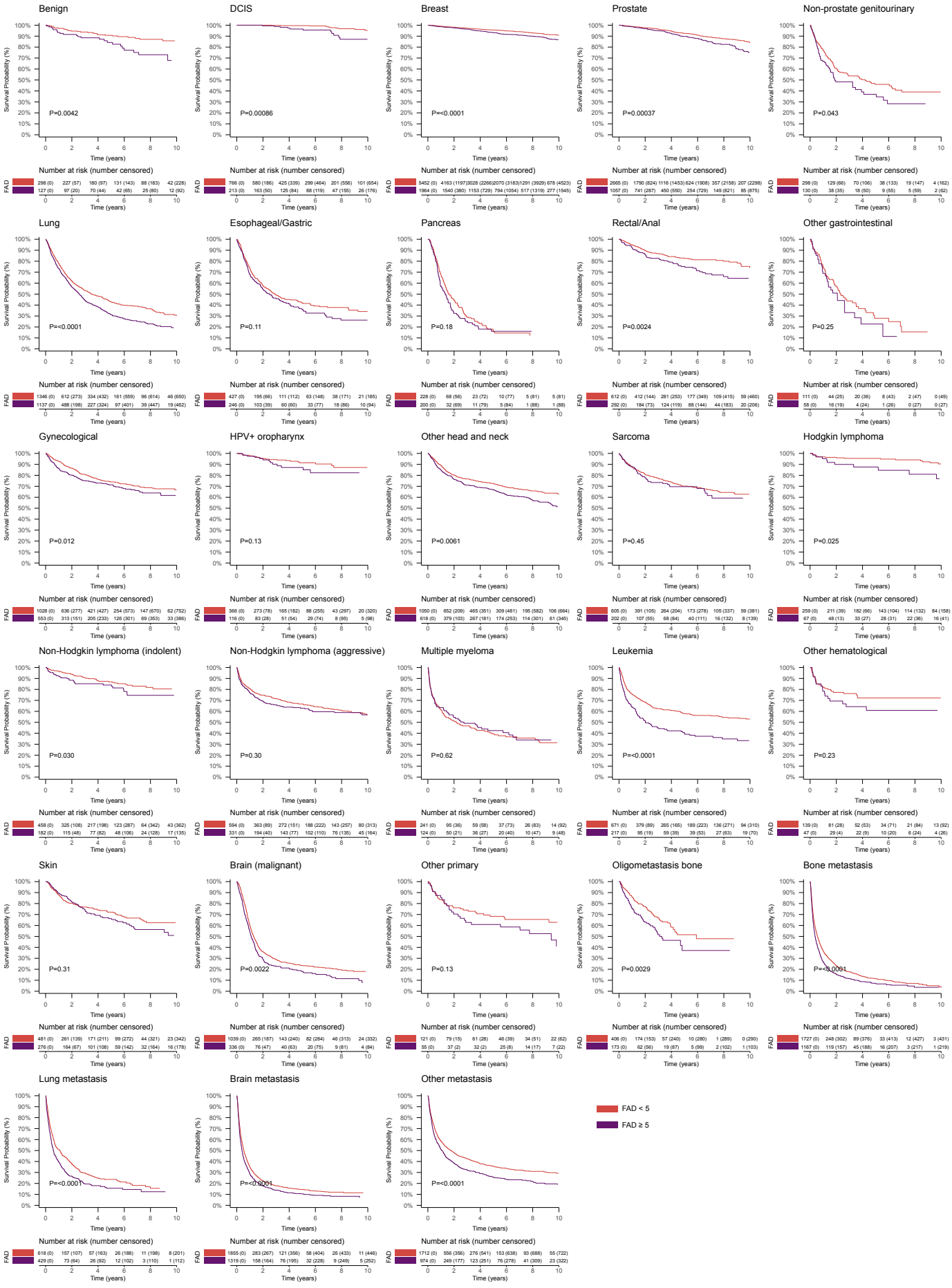
**d,** MAE for the Harvard RT patient dataset after the same fine-tuning, shown as a bar plot. The line plot displays the performance for each 5-year age bin for the top five models. The models included are FaceAge, FAHR-FaceAge, SENet50 IMDB-WIKI, DenseNet IMDB-WIKI and SENet50 VMAGE.

**e**, Analysis of plastic surgery procedures' effects on age estimation. Box plots compare pre- and post-surgery age estimates for five procedures: eyebrow correction (n=126 pairs, mean difference=0.29 years), eyelid correction (n=130 pairs, mean difference=-0.45 years), facelift (n=96 pairs, mean difference=-0.40 years), facial bones correction (n=105 pairs, mean difference=1.03 years), and nose correction (n=174 pairs, mean difference=1.11 years). Statistical analysis revealed significant changes only for nose correction ($P<0.0001$), while other procedures showed non-significant differences (all $P>0.05$).

**f**, Assessment of makeup effects on age estimation (n=26 pairs). Box plots display age estimates before and after makeup application. Analysis showed non-significant changes in age estimation (mean difference=1.09 years, $P>0.05$).

**g**, Evaluation of viewing conditions effects. Box plots show age estimates across different presentation variables: viewing angles (22.5° left vs. frontal: mean difference=0.25 years; 22.5° right vs. frontal: mean difference=0.55 years; n=200 pairs), facial expressions (neutral vs. smiling: mean difference=0.79 years, n=200 pairs), and illumination conditions (light vs. dark: mean difference=0.55 years, n=181 pairs). All variations produced non-significant differences (all $P>0.05$).

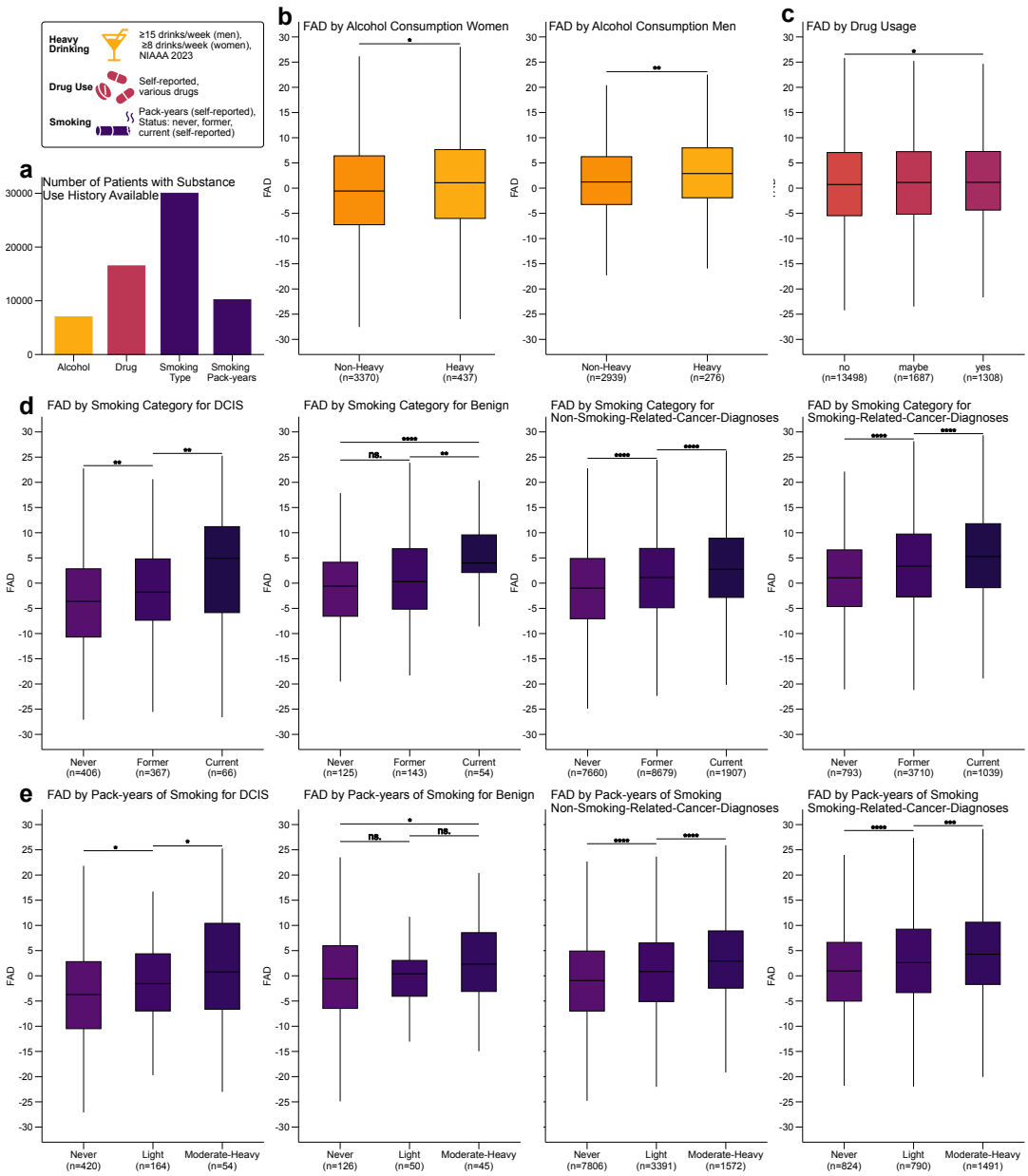
(**e,f,g**), The Wilcoxon signed-rank test was used to determine significance for each paired comparison, with the following annotation: ns $P>0.05$, *$P<0.05$, **$P<0.01$, ***$P<0.001$, ****$P<0.0001$.




**Extended Data Figure 2: Kaplan-Meier Survival Curves by Cancer Diagnosis with FAD≥5 and FAD<5 on the Harvard RT Patient Dataset**

Kaplan-Meier survival curves for each cancer diagnosis, stratified by FAHR-FaceAge deviation (FAD) with the split at FAD≥5 and FAD<5. The plots demonstrate the survival probabilities over time for patients within each cancer

diagnosis based on their FAD value. For each cancer diagnosis, the survival curves illustrate that patients with an FAD≥5 generally exhibit lower survival probabilities compared to those with an FAD<5. The significance of the difference between the groups is assessed using the log-rank test (P-values). DCIS, ductal carcinoma in situ; HPV, human papillomavirus.




**Extended Data Figure 3: Impact of Lifestyle Factors on FAHR-FaceAge Deviation (FAD) on the Harvard RT Patient Dataset**

**a,** Bar plot showing the number of patients with self-reported data on smoking, alcohol consumption, and drug use. This panel illustrates the amount of available data for each lifestyle factor.

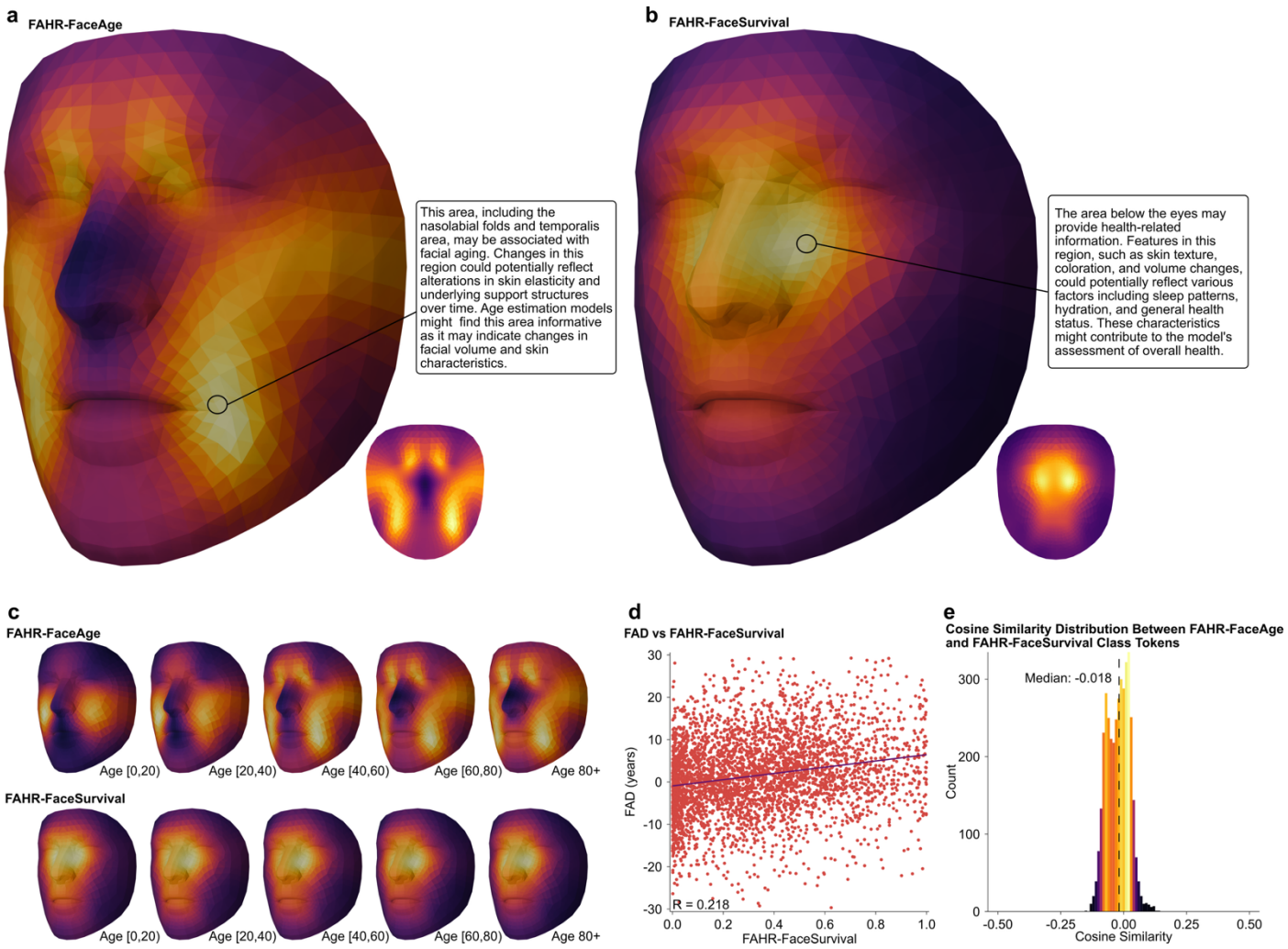
**b,** Boxplots showing FAD grouped by heavy alcohol use, defined by NIAAA 2023 guidelines as consuming 15 or more drinks per week for men and 8 or more drinks per week for women. The plots show the distribution of FAD for heavy drinkers versus non-heavy drinkers for both men and women.

**c,** Boxplots showing FAD grouped by self-reported drug use categorized into three groups: No, Maybe, and Yes. This panel shows the distribution of FAD based on drug use.

**d,** Boxplots showing FAD grouped by self-reported smoking status: Never, Former, and Current.

**e,** Boxplots of FAD grouped by self-reported pack-years of cigarette consumption into never**,** light and moderate to heavy smokers. The definition of smoking-related and non-smoking-related cancers is as follows: Smoking-related cancers include Lung, Other HN, Non-prostate GU, and Pancreas. Non-smoking-related cancers include Breast, Prostate, Skin, HPV+ OP, NHL Indolent, HL (Hodgkin Lymphoma), Rectal/Anal, NHL Aggressive, Leukemia, Myeloma, Other Heme, Gyn, Sarcoma, Other GI, Other primary, Esophageal/Gastric, Brain Malignant, Brain, Hodgkin, Rectal/Anus, NHL (Aggressive), Other heme, NHL (Indolent), and GYN. Metastatic cancers were excluded. A relative risk of ≥3 was used to define smoking-related cancers[26]. The pack-year groups are defined as follows: "Never" (0 pack-years), "Light" (0 < pack-years ≤ 20), "Moderate" (20 < pack-years ≤ 40), and "Heavy" (pack-years > 40)[27].

(**b,c,d,e**), The Wilcoxon rank-sum test was used to determine significance for each comparison, with the following annotation for significance: ns (P>0.05), *P<0.05, **P<0.01, ***P<0.001, ****P<0.0001. Boxplots show median and IQR; whiskers indicate 1.5×IQR.



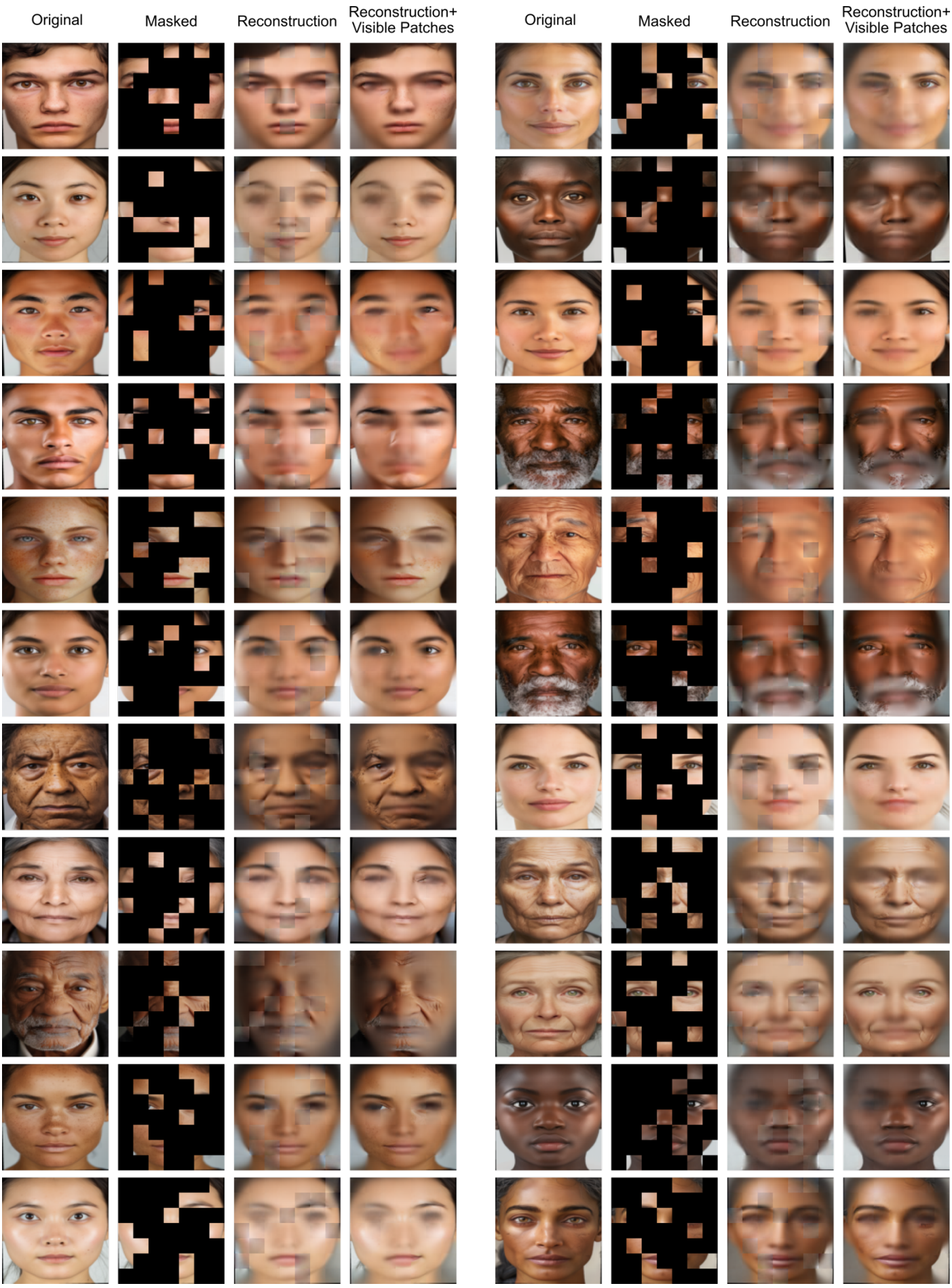

**Extended Data Figure 4: Mean Attention Maps and Correlation Analysis for FAHR-FaceAge and FAHR-FaceSurvival Models on the Harvard RT Patient Test Dataset**

**a,b,** Mean Attention maps projected onto 3D and 2D canonical face models for FAHR-FaceAge (a) and FAHR-FaceSurvival (b) for Harvard RT patient test dataset (n=3,815). The heat maps highlight areas of focus for each model's predictive task. FAHR-FaceAge concentrates on regions associated with age-related facial changes particular on nasolabial folds and temporalis area. FAHR-FaceSurvival emphasizes features under the eyes.

**c,** Mean Age-stratified attention maps for both FAHR-FaceAge and FAHR-FaceSurvival models for Harvard RT patient test dataset (n=3,815).

**d,** Scatter plot showing the relationship between FAD and FAHR-FaceSurvival. The plot reveals a weak positive correlation (R = 0.218) between the two biomarkers. FAD, FAHR-FaceAge deviation (FAHR-FaceAge minus chronological age).

**e,** Distribution of cosine similarities between FAHR-FaceAge and FAHR-FaceSurvival class token embeddings. The median similarity is -0.018.




**Extended Data Figure 5: Reconstruction Ability of the Masked Autoencoder FAHR-Face Foundation**

This figure demonstrates the reconstruction ability of the masked autoencoder (MAE). Synthetic faces were generated using Midjourney with the following prompt: "Create a realistic face photo of a person, with the following attributes randomly selected: race, age, and sex. The individual can be of any ethnic background, any age from a child to an elderly person, and any sexes. The face should have natural expressions and be set in different lighting conditions and backgrounds. Ensure only one person is in the photo, and the appearance should be realistic and natural." All photos that contained exactly one face were used.

The images show the original synthetic faces and their reconstructions by the masked autoencoder across various ages, sex, and ethnic backgrounds. The reconstructions demonstrate the masked autoencoder's ability to accurately capture and recreate key facial features and structures, maintaining the integrity of the original faces. The performance of the masked autoencoder highlights its robustness and generalizability across diverse demographic characteristics.




**Extended Data Figure 6: Age Dataset Collection, Distribution, and Oversampling**

**a,** Flowchart illustrating the collection and preprocessing of public datasets with age labels. The datasets include IMDB-WIKI, FG-NET, CACD, MegaAge, MORPH, Lagenda, KANFace, and AFAD for pre-training; UTK and AgeDB for fine-tuning; and APPA-Real for evaluation. The preprocessing steps involve cropping and aligning photos where exactly one face is detected, and filtering individuals aged between 1 and 116 years. The final number of photos used from each dataset is shown.

**b,** Age distribution of all used datasets. Each subplot shows the age distribution for a specific dataset used in pre-training (IMDB-WIKI, FG-NET, CACD, MegaAge, MORPH, Lagenda, KANFace, AFAD), fine-tuning (UTK, AgeDB), and evaluation (APPA-Real). The distributions illustrate the range of ages covered by each dataset, highlighting the diversity and representation across different age groups.

**c,** Illustration of the oversampling strategy applied to the datasets. The left panel shows the age distribution for the pre-training datasets before and after oversampling, while the right panel shows the same for the fine-tuning datasets. Oversampling is used to balance the age distribution, ensuring sufficient representation across all age groups and enhancing the model's ability to generalize.

**Extended Data Table 1. Patient demographics and characteristics of Harvard RT Patient Dataset and Maastro RT Patient Dataset**

| Characteristics | Harvard RT Patient Dataset | | | Maastro RT Patient Dataset (n=4,892) |
|---|---|---|---|---|
| | FAHR-FaceAge (N=38,183) | FAHR-FaceSurvival (N=34,363) - training | FAHR-FaceSurvival (N=3,820) - test | |
| Age (years), median (range) | 64.1 (0.7-102.7) | 64.1 (0.7-102.7) | 64.1 (0.8-98.5) | 66.0 (22.0-94.0) |
| Number of events (deaths), No. (%) | 13,769 (36.1) | 12,357 (36.0) | 1,412 (37.0) | 1811 (37.0) |
| Median follow-up time[a] (years) | 4.3 | 4.3 | 4.2 | 4.9 |
| FAHR-FaceAge (years), median (range) | 65.2 (-1.7-97.3) | 65.2 (-1.7-97.3) | 65.4 (-0.5-97.3) | 70.9 (12.0 to 97.0) |
| *FAD* (years) | | | | |
| Median (range) | 1.1 (-63.4-73.5) | 1.1 (-63.4-73.5) | 1.3 (-58.1-60.5) | 4.4 (-31.0 to 39.6) |
| ≥5, No. (%) | 12,610 (33.0) | 11,337 (33.0) | 1,273 (33.3) | 2,288 (46.8) |
| ≥10, No. (%) | 6,506 (17.0) | 5,828 (17.0) | 678 (17.7) | 1,103 (22.5) |
| ≤-5, No. (%) | 9,267 (24.3) | 8,413 (24.5) | 854 (22.4) | 410 (8.4%) |
| FAHR-FaceSurvival | | | | |
| Median (range) | - | - | 0.3 (0.0-1.0) | 0.4 (0.0-1.0) |
| <0.25 | - | - | 1,685 (44.1) | 1,405 (28.7) |
| 0.25-0.49 | - | - | 1,124 (29.4) | 1,703 (34.8) |
| 0.5-0.74 | - | - | 672 (17.6) | 1,461 (29.9) |
| ≥0.75 | - | - | 339 (8.9) | 323 (6.6) |
| Sex, No. (%) | | | | |
| Female | 21,557 (56.5) | 19,379 (56.4) | 2,178 (57.0) | 2,309 (47.2) |
| Male | 16,626 (43.5) | 14,984 (43.6) | 1,642 (43.0) | 2,583 (52.8) |
| Race/ethnicity, No. (%) | | | | |
| White | 32,553 (85.3) | 29,298 (85.3) | 3,255 (85.2) | - |
| Black | 1,821 (4.8) | 1,639 (4.8) | 182 (4.8) | - |
| Asian | 1,239 (3.2) | 1,123 (3.3) | 116 (3.0) | - |
| Hispanic | 609 (1.6) | 550 (1.6) | 59 (1.5) | - |
| Other | 978 (2.6) | 872 (2.5) | 106 (2.8) | - |
| Unknown | 983 (2.6) | 881 (2.6) | 102 (2.7) | - |
| Cancer risk group, No. (%) | | | | |
| Benign/DCIS | 1,404 (3.7) | 1,259 (3.7) | 145 (3.8) | - |
| Non-metastatic cancer | 26,399 (69.1) | 23,769 (69.2) | 2,630 (68.8) | - |
| Metastatic cancer | 10,380 (27.2) | 9,335 (27.2) | 1,045 (27.4) | - |
| Diagnosis, No. (%) | | | | |
| Benign | 425 (1.1) | 378 (1.1) | 47 (1.2) | - |
| DCIS | 979 (2.6) | 881 (2.6) | 98 (2.6) | - |
| Breast | 7,416 (19.4) | 6,709 (19.5) | 707 (18.5) | - |
| Prostate | 3,722 (9.7) | 3,341 (9.7) | 381 (10.0) | - |
| Non-prostate genitourinary | 428 (1.1) | 398 (1.2) | 30 (0.8) | - |
| Lung | 2,483 (6.5) | 2,241 (6.5) | 242 (6.3) | - |
| Esophageal/Gastric | 673 (1.8) | 613 (1.8) | 60 (1.6) | - |
| Pancreas | 428 (1.1) | 388 (1.1) | 40 (1.0) | - |
| Rectal/Anal | 904 (2.4) | 812 (2.4) | 92 (2.4) | - |
| Other gastrointestinal | 169 (0.4) | 150 (0.4) | 19 (0.5) | - |
| Gynecological | 1,581 (4.1) | 1,426 (4.1) | 155 (4.1) | - |
| HPV+ oropharynx | 482 (1.3) | 442 (1.3) | 40 (1.0) | - |
| Other head and neck | 1,668 (4.4) | 1,476 (4.3) | 192 (5.0) | - |
| Sarcoma | 807 (2.1) | 727 (2.1) | 80 (2.1) | - |
| Hodgkin lymphoma | 326 (0.9) | 300 (0.9) | 26 (0.7) | - |
| Non-Hodgkin lymphoma (indolent) | 640 (1.7) | 574 (1.7) | 66 (1.7) | - |
| Non-Hodgkin lymphoma (aggressive) | 925 (2.4) | 810 (2.4) | 115 (3.0) | - |
| Multiple myeloma | 365 (1.0) | 331 (1.0) | 34 (0.9) | - |
| Leukemia | 888 (2.3) | 796 (2.3) | 92 (2.4) | - |
| Other hematological | 186 (0.5) | 169 (0.5) | 17 (0.4) | - |
| Skin | 757 (2.0) | 677 (2.0) | 80 (2.1) | - |
| Brain (malignant) | 1,375 (3.6) | 1,234 (3.6) | 141 (3.7) | - |
| Other primary | 176 (0.5) | 155 (0.5) | 21 (0.5) | - |
| Oligometastasis bone | 579 (1.5) | 546 (1.6) | 33 (0.9) | - |
| Bone metastasis | 2,894 (7.6) | 2,615 (7.6) | 279 (7.3) | - |
| Lung metastasis | 1,047 (2.7) | 938 (2.7) | 109 (2.9) | - |
| Brain metastasis | 3,174 (8.3) | 2,832 (8.2) | 342 (9.0) | - |
| Other metastasis | 2,686 (7.0) | 2,404 (7.0) | 282 (7.4) | - |
| Cancer site, No. (%) | | | | |
| Benign | 425 (1.1) | 378 (1.1) | 47 (1.2) | - |
| Breast | 8,395 (22.0) | 7,590 (22.1) | 805 (21.1) | - |
| Prostate | 3,722 (9.7) | 3,341 (9.7) | 381 (10.0) | - |
| Lung | 2,483 (6.5) | 2,241 (6.5) | 242 (6.3) | - |
| Gastrointestinal | 2,174 (5.7) | 1,963 (5.7) | 211 (5.5) | - |
| Gynecological | 1,581 (4.1) | 1,426 (4.1) | 155 (4.1) | - |
| Head and neck | 2,150 (5.6) | 1,918 (5.6) | 232 (6.1) | - |
| Sarcoma | 807 (2.1) | 727 (2.1) | 80 (2.1) | - |
| Hematological | 3,330 (8.7) | 2,980 (8.7) | 350 (9.2) | - |
| Skin | 757 (2.0) | 677 (2.0) | 80 (2.1) | - |
| Brain (malignant) | 1,375 (3.6) | 1,234 (3.6) | 141 (3.7) | - |
| Other primary | 604 (1.6) | 553 (1.6) | 51 (1.3) | - |

| | | | | |
|---|---|---|---|---|
| Bone metastasis | 3,473 (9.1) | 3,161 (9.2) | 312 (8.2) | - |
| Brain metastasis | 3,174 (8.3) | 2,832 (8.2) | 342 (9.0) | - |
| Other metastasis | 3,733 (9.8) | 3,342 (9.7) | 391 (10.2) | |
| Cancer site (Maastro), No. (%) | | | | |
| Breast | - | - | - | 1337 (27.3) |
| Lung | - | - | - | 734 (15) |
| Gastrointestinal | - | - | - | 999 (20.4) |
| Genitourinary | - | - | - | 838 (17.1) |
| Head and Neck | - | - | - | 455 (9.3) |
| Other | - | - | - | 529 (10.8) |
| Treatment course intent, No. (%) | | | | |
| Curative | 26,348 (69.0) | 23,752 (69.1) | 2,596 (68.0) | - |
| Oligometastasis ablation | 3,262 (8.5) | 2,953 (8.6) | 309 (8.1) | - |
| Palliative | 8,573 (22.5) | 7,658 (22.3) | 915 (24.0) | - |
| Year treated, No. (%) | | | | |
| 2008-2015 | 13,447 (35.2) | 12,112 (35.2) | 1,335 (34.9) | - |
| 2016-2023 | 24,736 (64.8) | 22,251 (64.8) | 2,485 (65.1) | - |
| Radiation technique, No. (%) | | | | |
| 3D | 18,866 (49.4) | 16,991 (49.4) | 1,875 (49.1) | - |
| IMRT | 10,366 (27.1) | 9,304 (27.1) | 1,062 (27.8) | - |
| SBRT | 1,787 (4.7) | 1,640 (4.8) | 147 (3.8) | - |
| SRS | 1,981 (5.2) | 1,768 (5.1) | 213 (5.6) | - |
| Brachytherapy | 1,292 (3.4) | 1,174 (3.4) | 118 (3.1) | - |
| Electrons | 864 (2.3) | 767 (2.2) | 97 (2.5) | - |
| Adaptive | 550 (1.4) | 486 (1.4) | 64 (1.7) | - |
| Unknown | 2,477 (6.5) | 2,233 (6.5) | 244 (6.4) | - |

Abbreviations: FAD, FAHR-FaceAge deviation (FAHR-FaceAge minus chronological age); No. (%), number of patients (percentage); DCIS, ductal carcinoma in situ; HPV, human papillomavirus; RT, radiotherapy; 3D, three-dimensional conformal radiotherapy; IMRT, intensity-modulated radiotherapy; SBRT, stereotactic body radiotherapy; SRS, stereotactic radiosurgery.

[a]Median follow-up time was calculated using the reverse Kaplan-Meier estimator by reversing the censoring indicator.

**Extended Data Table 2. Results for FAHR-FaceAge variables in multivariate Cox regression for survival within each subgroup by sex, race/ethnicity, cancer risk, prognosis group, cancer site, and treatment course intent**

| Subgroups for individual multivariate model | Univariate result for *FAD* [decade] | | Multivariate result for *FAD* [decade] | | Univariate result for *FAD* ≥5 years | | Multivariate result for *FAD* ≥5 years | | Univariate result for *FAD* ≤-5 years | | Multivariate result for *FAD* ≤-5 years | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | HR (95% CI) | P-value | HR (95% CI) | P- value | HR (95% CI) | P- value | HR (95% CI) | P- value | HR (95% CI) | P- value | HR (95% CI) | P-value |
| Sex[a] | | | | | | | | | | | | |
| Female | 1.35 (1.32-1.38) | <0.0001 | 1.16 (1.13-1.19) | <0.0001 | 1.76 (1.68-1.85) | <0.0001 | 1.26 (1.20-1.32) | <0.0001 | 0.61 (0.57-0.65) | <0.0001 | 0.78 (0.73-0.83) | <0.0001 |
| Male | 1.19 (1.15-1.22) | <0.0001 | 1.11 (1.07-1.14) | <0.0001 | 1.32 (1.25-1.39) | <0.0001 | 1.17 (1.11-1.23) | <0.0001 | 0.89 (0.83-0.95) | 0.0004 | 0.89 (0.83-0.95) | 0.00054 |
| Age[b] | | | | | | | | | | | | |
| <40 | 1.12 (1.03-1.22) | 0.0098 | 1.12 (1.02-1.23) | 0.02 | 1.33 (1.11-1.59) | 0.0022 | 1.45 (1.21-1.74) | <0.0001 | 0.90 (0.77-1.05) | 0.19 | 0.98 (0.82-1.16) | 0.80 |
| 40-60 | 1.29 (1.25-1.34) | <0.0001 | 1.17 (1.13-1.21) | <0.0001 | 1.53 (1.43-1.64) | <0.0001 | 1.26 (1.18-1.36) | <0.0001 | 0.64 (0.60-0.70) | <0.0001 | 0.79 (0.72-0.85) | <0.0001 |
| 60-80 | 1.29 (1.25-1.32) | <0.0001 | 1.15 (1.12-1.18) | <0.0001 | 1.49 (1.42-1.56) | <0.0001 | 1.25 (1.19-1.31) | <0.0001 | 0.72 (0.67-0.77) | <0.0001 | 0.83 (0.78-0.89) | <0.0001 |
| ≥80 | 1.14 (1.07-1.21) | <0.0001 | 1.14 (1.07-1.21) | <0.0001 | 1.11 (1.01-1.22) | 0.030 | 1.12 (1.01-1.24) | 0.025 | 0.79 (0.70-0.90) | 0.00051 | 0.78 (0.68-0.89) | 0.00028 |
| Race/ethnicity[c] | | | | | | | | | | | | |
| White | 1.31 (1.28-1.34) | <0.0001 | 1.15 (1.13-1.18) | <0.0001 | 1.54 (1.48-1.59) | <0.0001 | 1.23 (1.19-1.28) | <0.0001 | 0.68 (0.65-0.72) | <0.0001 | 0.83 (0.79-0.87) | <0.0001 |
| Black | 1.13 (1.06-1.22) | 0.00065 | 1.01 (0.94-1.08) | 0.86 | 1.31 (1.09-1.57) | 0.0032 | 0.94 (0.78-1.14) | 0.54 | 0.85 (0.72-1.01) | 0.065 | 0.95 (0.80-1.13) | 0.57 |
| Asian | 1.29 (1.18-1.40) | <0.0001 | 1.18 (1.07-1.30) | 0.001 | 1.73 (1.36-2.21) | <0.0001 | 1.32 (1.01-1.71) | 0.041 | 0.50 (0.40-0.62) | <0.0001 | 0.61 (0.48-0.77) | <0.0001 |
| Hispanic* | 1.15 (1.01-1.32) | 0.041 | 1.13 (0.98-1.31) | 0.093 | 1.33 (1.00-1.75) | 0.047 | 1.29 (0.96-1.73) | 0.093 | 0.59 (0.43-0.81) | 0.0011 | 0.68 (0.48-0.95) | 0.026 |
| Other | 1.22 (1.10-1.35) | 0.0002 | 1.05 (0.94-1.16) | 0.42 | 1.34 (1.06-1.70) | 0.015 | 1.10 (0.86-1.41) | 0.46 | 0.67 (0.52-0.87) | 0.0026 | 0.82 (0.63-1.07) | 0.15 |
| Cancer risk groups[d] | | | | | | | | | | | | |
| Benign/DCIS* | 1.69 (1.35-2.12) | <0.0001 | 1.39 (1.09-1.78) | 0.0091 | 2.91 (1.85-4.58) | <0.0001 | 1.94 (1.21-3.12) | 0.0062 | 0.42 (0.23-0.77) | 0.0046 | 0.66 (0.36-1.21) | 0.18 |
| Non-metastatic cancer | 1.28 (1.24-1.31) | <0.0001 | 1.18 (1.15-1.22) | <0.0001 | 1.46 (1.38-1.54) | <0.0001 | 1.29 (1.22-1.36) | <0.0001 | 0.64 (0.60-0.69) | <0.0001 | 0.75 (0.70-0.80) | <0.0001 |
| Metastatic cancer | 1.16 (1.14-1.19) | <0.0001 | 1.13 (1.11-1.16) | <0.0001 | 1.31 (1.25-1.38) | <0.0001 | 1.21 (1.15-1.27) | <0.0001 | 0.83 (0.78-0.88) | <0.0001 | 0.85 (0.80-0.90) | <0.0001 |
| Cancer site[e] | | | | | | | | | | | | |
| Breast | 1.40 (1.27-1.54) | <0.0001 | 1.28 (1.15-1.42) | <0.0001 | 1.65 (1.35-2.02) | <0.0001 | 1.37 (1.12-1.68) | 0.0025 | 0.51 (0.40-0.65) | <0.0001 | 0.62 (0.48-0.80) | 0.00021 |
| Prostate | 1.35 (1.14-1.60) | 0.00045 | 1.27 (1.08-1.50) | 0.0044 | 1.50 (1.15-1.95) | 0.0026 | 1.42 (1.09-1.86) | 0.0092 | 0.80 (0.56-1.14) | 0.22 | 0.78 (0.54-1.12) | 0.18 |
| Lung | 1.17 (1.10-1.24) | <0.0001 | 1.17 (1.10-1.25) | <0.0001 | 1.29 (1.15-1.45) | <0.0001 | 1.34 (1.19-1.51) | <0.0001 | 0.69 (0.58-0.82) | <0.0001 | 0.78 (0.66-0.93) | 0.0046 |
| Gastrointestinal | 1.24 (1.15-1.34) | <0.0001 | 1.14 (1.05-1.23) | 0.002 | 1.34 (1.16-1.54) | <0.0001 | 1.10 (0.95-1.27) | 0.22 | 0.63 (0.52-0.76) | <0.0001 | 0.75 (0.62-0.90) | 0.0027 |
| Gynecological | 1.11 (1.00-1.24) | 0.043 | 1.05 (0.95-1.17) | 0.36 | 1.37 (1.09-1.71) | 0.0066 | 1.17 (0.93-1.48) | 0.17 | 0.86 (0.66-1.10) | 0.23 | 0.87 (0.67-1.12) | 0.28 |
| Head and neck | 1.21 (1.10-1.34) | 0.00021 | 1.13 (1.02-1.25) | 0.019 | 1.44 (1.20-1.72) | <0.0001 | 1.30 (1.08-1.57) | 0.0052 | 0.83 (0.65-1.05) | 0.12 | 0.84 (0.66-1.08) | 0.17 |
| Sarcoma | 1.03 (0.89-1.20) | 0.67 | 0.96 (0.83-1.11) | 0.55 | 1.13 (0.83-1.54) | 0.45 | 0.87 (0.62-1.22) | 0.42 | 1.08 (0.80-1.46) | 0.63 | 1.16 (0.85-1.60) | 0.35 |
| Hematological | 1.17 (1.09-1.26) | <0.0001 | 1.11 (1.04-1.20) | 0.0032 | 1.40 (1.23-1.59) | <0.0001 | 1.22 (1.07-1.40) | 0.0034 | 0.89 (0.76-1.05) | 0.16 | 0.88 (0.75-1.03) | 0.12 |
| Skin | 1.19 (1.00-1.42) | 0.055 | 1.42 (1.15-1.74) | 0.00093 | 1.26 (0.94-1.70) | 0.12 | 1.61 (1.17-2.21) | 0.0032 | 0.68 (0.44-1.05) | 0.085 | 0.62 (0.39-0.98) | 0.039 |
| Brain (malignant) | 1.16 (1.08-1.26) | 0.00011 | 1.12 (1.04-1.21) | 0.0049 | 1.22 (1.05-1.42) | 0.0088 | 1.11 (0.95-1.29) | 0.20 | 0.77 (0.65-0.90) | 0.0011 | 0.79 (0.67-0.93) | 0.0039 |
| Treatment course intent[f] | | | | | | | | | | | | |
| Curative | 1.31 (1.27-1.35) | <0.0001 | 1.16 (1.12-1.20) | <0.0001 | 1.53 (1.45-1.62) | <0.0001 | 1.24 (1.17-1.32) | <0.0001 | 0.62 (0.57-0.66) | <0.0001 | 0.78 (0.73-0.84) | <0.0001 |
| Oligomet. ablation | 1.21 (1.15-1.27) | <0.0001 | 1.15 (1.09-1.21) | <0.0001 | 1.32 (1.20-1.46) | <0.0001 | 1.19 (1.07-1.32) | 0.0011 | 0.81 (0.71-0.91) | 0.0004 | 0.84 (0.74-0.95) | 0.0048 |
| Palliative | 1.13 (1.10-1.16) | <0.0001 | 1.11 (1.09-1.14) | <0.0001 | 1.23 (1.17-1.29) | <0.0001 | 1.19 (1.13-1.25) | <0.0001 | 0.82 (0.77-0.87) | <0.0001 | 0.85 (0.80-0.91) | <0.0001 |

Abbreviations: FAD, FAHR-FaceAge deviation (FAHR-FaceAge minus chronological Age); HR, hazard ratio; CI, confidence interval; DCIS, ductal carcinoma in situ; oligomet., oligometastasis.

[a]Multivariate analysis with age, race, cancer site, treatment course intent, year treated, radiation technique incorporated

[b]Multivariate analysis with sex, race, cancer site, treatment course intent, year treated, radiation technique incorporated

[c]Multivariate analysis with age, sex, cancer site, treatment course intent, year treated, radiation technique incorporated; *multivariate analysis with age, sex, cancer site, treatment course intent, year treated only for Hispanic to avoid overfitting in setting of limited number of events

[d]Multivariate analysis with age, sex, race, treatment course intent, year treated, radiation technique incorporated; *multivariate analysis with age, sex, treatment course intent only for benign/DCIS to avoid overfitting in setting of limited number of events

[e]Multivariate analysis with age, sex, race, treatment course intent, year treated, radiation technique incorporated
[f]Multivariate analysis with age, sex, race, cancer site, year treated, radiation technique incorporated

**Extended Data Table 3. Results for FAHR-FaceSurvival in univariate and multivariate Cox regression for overall survival within each subgroup by sex, race/ethnicity, cancer risk, prognosis group, cancer site, and treatment course intent for the Harvard RT test patient dataset**

| Subgroups | Univariate for FAHR-FaceSurvival (0.1) | | Multivariate result for FAHR-FaceSurvival (0.1) | | Univariate result for FAHR-FaceSurvival (≥0.5vs<0.5) | | Multivariate result for FAHR-FaceSurvival (≥0.5vs<0.5) | |
|---|---|---|---|---|---|---|---|---|
| | HR (95% CI) | P-value | HR (95% CI) | P-value | HR (95% CI) | P-value | HR (95% CI) | P-value |
| Sex[a] | | | | | | | | |
| Female | 1.35 (1.32-1.39) | <0.0001 | 1.20 (1.14-1.25) | <0.0001 | 4.47 (3.86-5.17) | <0.0001 | 2.22 (1.72-2.87) | <0.0001 |
| Male | 1.28 (1.24-1.32) | <0.0001 | 1.19 (1.13-1.25) | <0.0001 | 2.89 (2.48-3.37) | <0.0001 | 2.02 (1.57-2.59) | <0.0001 |
| Age[b] | | | | | | | | |
| <40* | 1.18 (1.10-1.27) | <0.0001 | 1.11 (1.02-1.20) | 0.019 | 1.98 (1.31-2.98) | 0.0011 | 1.20 (0.72-2.01) | 0.49 |
| 40-60 | 1.35 (1.30-1.39) | <0.0001 | 1.17 (1.10-1.25) | <0.0001 | 5.12 (4.16-6.31) | <0.0001 | 2.46 (1.67-3.62) | <0.0001 |
| 60-80 | 1.34 (1.31-1.38) | <0.0001 | 1.26 (1.20-1.33) | <0.0001 | 3.51 (3.03-4.06) | <0.0001 | 2.18 (1.71-2.80) | <0.0001 |
| ≥80* | 1.21 (1.13-1.29) | <0.0001 | 1.15 (1.07-1.23) | <0.0001 | 2.28 (1.70-3.08) | <0.0001 | 2.05 (1.49-2.80) | <0.0001 |
| Race/ethnicity[c] | | | | | | | | |
| White | 1.33 (1.30-1.36) | <0.0001 | 1.20 (1.16-1.25) | <0.0001 | 3.82 (3.41-4.28) | <0.0001 | 2.25 (1.87-2.71) | <0.0001 |
| Black* | 1.41 (1.28-1.56) | <0.0001 | 1.27 (1.14-1.42) | <0.0001 | 5.38 (3.13-9.24) | <0.0001 | 3.59 (1.98-6.51) | <0.0001 |
| Asian* | 1.32 (1.17-1.49) | <0.0001 | 1.29 (1.14-1.47) | <0.0001 | 4.44 (2.17-9.09) | <0.0001 | 3.79 (1.74-8.28) | 0.00082 |
| Hispanic* | 1.14 (0.97-1.35) | 0.11 | 1.18 (0.99-1.41) | 0.060 | 1.91 (0.78-4.69) | 0.16 | 2.09 (0.84-5.24) | 0.11 |
| Other* | 1.29 (1.15-1.44) | <0.0001 | 1.20 (1.06-1.35) | 0.0030 | 2.23 (1.24-4.00) | 0.0071 | 1.93 (1.03-3.59) | 0.039 |
| Cancer risk groups[d] | | | | | | | | |
| Benign/DCIS* | 1.17 (0.99-1.39) | 0.072 | -- | -- | 2.83 (0.73-10.97) | 0.13 | -- | -- |
| Non-metastatic cancer | 1.33 (1.29-1.37) | <0.0001 | 1.26 (1.22-1.30) | <0.0001 | 3.62 (3.10-4.23) | <0.0001 | 2.75 (2.32-3.26) | <0.0001 |
| Metastatic cancer | 1.19 (1.15-1.22) | <0.0001 | 1.16 (1.12-1.19) | <0.0001 | 2.19 (1.89-2.54) | <0.0001 | 1.88 (1.61-2.19) | <0.0001 |
| Cancer site[e] | | | | | | | | |
| Breast* | 1.31 (1.16-1.49) | <0.0001 | 1.26 (1.10-1.43) | 0.00055 | 3.86 (1.64-9.06) | 0.0019 | 3.02 (1.27-7.16) | 0.012 |
| Prostate* | 1.46 (1.20-1.76) | 0.00013 | 1.41 (1.15-1.72) | 0.00075 | 4.03 (1.90-8.55) | 0.00028 | 3.80 (1.68-8.61) | 0.0014 |
| Lung* | 1.18 (1.09-1.29) | <0.0001 | 1.19 (1.08-1.30) | 0.00019 | 1.76 (1.23-2.51) | 0.0019 | 1.76 (1.20-2.59) | 0.004 |
| Gastrointestinal* | 1.20 (1.11-1.31) | <0.0001 | 1.11 (1.00-1.22) | 0.041 | 2.41 (1.54-3.78) | 0.00012 | 1.92 (1.18-3.11) | 0.0081 |
| Gynecological* | 1.21 (1.06-1.39) | 0.0063 | 1.08 (0.93-1.25) | 0.33 | 1.86 (0.85-4.08) | 0.12 | 0.83 (0.36-1.92) | 0.66 |
| Head and neck* | 1.27 (1.15-1.40) | <0.0001 | 1.21 (1.08-1.35) | 0.00072 | 3.39 (2.00-5.76) | <0.0001 | 2.57 (1.45-4.57) | 0.0013 |
| Sarcoma* | 1.20 (1.01-1.43) | 0.039 | 1.10 (0.92-1.32) | 0.29 | 3.48 (1.61-7.56) | 0.0016 | 3.48 (1.55-7.81) | 0.0025 |
| Hematological* | 1.29 (1.20-1.38) | <0.0001 | 1.29 (1.20-1.38) | <0.0001 | 2.31 (1.63-3.29) | <0.0001 | 2.48 (1.73-3.57) | <0.0001 |
| Skin* | 1.30 (1.05-1.61) | 0.016 | -- | -- | 2.73 (1.03-7.24) | 0.044 | -- | -- |
| Brain (malignant)* | 1.09 (1.02-1.17) | 0.012 | 1.08 (1.00-1.17) | 0.046 | 1.44 (0.96-2.16) | 0.082 | 1.27 (0.80-2.00) | 0.30 |
| Treatment course intent[f] | | | | | | | | |
| Curative | 1.27 (1.23-1.31) | <0.0001 | 1.14 (1.09-1.19) | <0.0001 | 2.95 (2.47-3.52) | <0.0001 | 1.73 (1.40-2.15) | <0.0001 |
| Oligomet. ablation* | 1.23 (1.16-1.31) | <0.0001 | 1.06 (0.87-1.30) | 0.56 | 3.06 (2.23-4.20) | <0.0001 | 1.40 (0.57-3.43) | 0.46 |
| Palliative | 1.18 (1.15-1.21) | <0.0001 | 1.28 (1.20-1.38) | <0.0001 | 2.08 (1.79-2.42) | <0.0001 | 3.02 (2.09-4.37) | <0.0001 |

Abbreviations: FAHR-FaceSurvival, Foundation Artificial Intelligence for Health Recognition survival score based on facial photographs; cont, continuous; cat, categorized; HR, hazard ratio; CI, confidence interval; DCIS, ductal carcinoma in situ; oligomet., oligometastasis

[a]Multivariate analysis with age, race, cancer site, treatment course intent, year treated, radiation technique incorporated

[b]Multivariate analysis with sex, race, cancer site, treatment course intent, year treated, radiation technique incorporated; *multivariate analysis with age, sex, race, treatment course intent only for age<40 and age≥80 to avoid overfitting in setting of limited number of events

[c]Multivariate analysis with age, sex, cancer site, treatment course intent, year treated, radiation technique incorporated; *multivariate analysis with age, sex, treatment course intent only for Black/others; age and sex only for Asian/Hispanic, to avoid overfitting in setting of limited number of events

[d]Multivariate analysis with age, sex, race, treatment course intent, year treated, radiation technique incorporated (*Benign/DCIS: no multivariate model due to limited number of events)

[e]Multivariate analysis with age, sex, race, treatment course intent, year treated, and radiation technique incorporated; site-specific reductions applied as follows: *Lung/Gastrointestinal/Hematological/Brain (malignant): age, sex, race, treatment course intent; Breast/Prostate/Gynecological/Sarcoma: age, treatment course intent; Head and neck: age, sex, treatment course intent; Skin: no multivariate model due to limited events.
[f]Multivariate analysis with age, sex, race, cancer site, year treated, radiation technique incorporated; *multivariate analysis with age, sex, and cancer site only for oligomet. ablation, to avoid overfitting in setting of limited number of events.

**Extended Data Table 4. Univariate and multivariate Cox regression for overall survival in Harvard RT patient dataset for FaceAge**[22]

| Cox regression | Univariate[a] | | Multivariate model 1[b] ([1]FaceAge minus age [decade]) | | Multivariate model 2[b] ([2]FaceAge minus age ≥5 years) | | Multivariate model 3[b] ([3]FaceAge minus age ≤-5 years) | |
|---|---|---|---|---|---|---|---|---|
| **Variables** | HR (95% CI) | P-value | HR (95% CI) | P-value | HR (95% CI) | P-value | HR (95% CI) | P-value |
| Age (decade) | 1.19 (1.18-1.21) | <0.0001 | 1.20 (1.18-1.21) | <0.0001 | 1.18 (1.16-1.19) | <0.0001 | 1.16 (1.14-1.17) | <0.0001 |
| FaceAge (decade) | 1.26 (1.24-1.28) | <0.0001 | -- | -- | -- | -- | -- | -- |
| [1]FaceAge minus age (decade) | 1.01 (0.99-1.03) | 0.28 | 1.15 (1.12-1.17) | <0.0001 | -- | -- | -- | -- |
| [2]FaceAge minus age ,≥5 vs <5 | 1.08 (1.04-1.11) | <0.0001 | -- | -- | 1.18 (1.14-1.23) | <0.0001 | -- | -- |
| [3]FaceAge minus age ,≤-5 vs >-5 | 1.05 (1.00-1.11) | 0.069 | -- | -- | -- | -- | 0.95 (0.90-1.01) | 0.083 |
| Sex, male vs female | 1.38 (1.34-1.43) | <0.0001 | 1.13 (1.09-1.17) | <0.0001 | 1.13 (1.09-1.17) | <0.0001 | 1.11 (1.07-1.15) | <0.0001 |
| Race/ethnicity[c] | -- | -- | -- | -- | -- | -- | -- | -- |
| White | Ref | -- | Ref | -- | Ref | -- | Ref | -- |
| Black | 0.90 (0.83-0.98) | 0.014 | 1.10 (1.01-1.20) | 0.027 | 1.08 (1.00-1.18) | 0.06 | 1.08 (0.99-1.17) | 0.087 |
| Asian | 0.83 (0.74-0.92) | 0.0004 | 0.99 (0.89-1.10) | 0.87 | 0.98 (0.88-1.09) | 0.73 | 0.97 (0.87-1.08) | 0.61 |
| Hispanic | 1.01 (0.88-1.15) | 0.94 | 1.20 (1.05-1.38) | 0.0076 | 1.20 (1.05-1.37) | 0.0091 | 1.19 (1.04-1.36) | 0.011 |
| Other | 0.96 (0.86-1.07) | 0.47 | 1.14 (1.02-1.27) | 0.026 | 1.14 (1.02-1.27) | 0.026 | 1.14 (1.02-1.28) | 0.024 |
| Cancer risk group[d] | -- | -- | -- | -- | -- | -- | -- | -- |
| Non-metastatic cancer | Ref | -- | -- | -- | -- | -- | -- | -- |
| Benign/DCIS | 0.18 (0.15-0.23) | <0.0001 | -- | -- | -- | -- | -- | -- |
| Metastatic cancer | 5.76 (5.57-5.97) | <0.0001 | -- | -- | -- | -- | -- | -- |
| Cancer site | -- | -- | -- | -- | -- | -- | -- | -- |
| Breast | Ref | -- | Ref | -- | Ref | -- | Ref | -- |
| Benign | 2.42 (1.82-3.21) | <0.0001 | 2.73 (2.04-3.66) | <0.0001 | 2.75 (2.05-3.69) | <0.0001 | 2.75 (2.05-3.69) | <0.0001 |
| Prostate | 1.67 (1.44-1.93) | <0.0001 | 1.75 (1.47-2.07) | <0.0001 | 1.76 (1.48-2.09) | <0.0001 | 1.77 (1.49-2.10) | <0.0001 |
| Lung | 15.40 (13.83-17.16) | <0.0001 | 13.98 (12.41-15.74) | <0.0001 | 14.13 (12.54-15.92) | <0.0001 | 14.34 (12.73-16.15) | <0.0001 |
| Gastrointestinal | 10.71 (9.56-12.01) | <0.0001 | 10.01 (8.85-11.33) | <0.0001 | 10.04 (8.87-11.35) | <0.0001 | 10.13 (8.95-11.46) | <0.0001 |
| Gynecological | 5.47 (4.75-6.29) | <0.0001 | 6.71 (5.76-7.83) | <0.0001 | 6.75 (5.79-7.87) | <0.0001 | 6.80 (5.83-7.93) | <0.0001 |
| Head and neck | 5.54 (4.88-6.28) | <0.0001 | 5.97 (5.18-6.89) | <0.0001 | 6.01 (5.21-6.93) | <0.0001 | 6.06 (5.26-7.00) | <0.0001 |
| Sarcoma | 6.26 (5.33-7.35) | <0.0001 | 6.95 (5.86-8.25) | <0.0001 | 6.97 (5.87-8.26) | <0.0001 | 6.98 (5.88-8.28) | <0.0001 |
| Hematological | 7.53 (6.75-8.41) | <0.0001 | 4.51 (3.99-5.09) | <0.0001 | 4.55 (4.02-5.14) | <0.0001 | 4.57 (4.04-5.17) | <0.0001 |
| Skin | 6.54 (5.54-7.72) | <0.0001 | 6.61 (5.44-8.02) | <0.0001 | 6.69 (5.51-8.12) | <0.0001 | 6.76 (5.57-8.21) | <0.0001 |
| Brain (malignant) | 24.57 (21.93-27.52) | <0.0001 | 25.44 (22.38-28.91) | <0.0001 | 25.34 (22.29-28.79) | <0.0001 | 25.27 (22.24-28.72) | <0.0001 |
| Other primary | 11.71 (10.05-13.64) | <0.0001 | 10.21 (8.65-12.06) | <0.0001 | 10.40 (8.81-12.28) | <0.0001 | 10.45 (8.85-12.33) | <0.0001 |
| Bone metastasis | 36.78 (33.21-40.73) | <0.0001 | 12.94 (11.46-14.61) | <0.0001 | 13.07 (11.58-14.76) | <0.0001 | 13.14 (11.64-14.84) | <0.0001 |
| Brain metastasis | 44.94 (40.58-49.77) | <0.0001 | 21.58 (19.03-24.48) | <0.0001 | 21.76 (19.19-24.68) | <0.0001 | 22.04 (19.44-25.00) | <0.0001 |
| Other metastasis | 24.59 (22.21-27.24) | <0.0001 | 13.16 (11.68-14.81) | <0.0001 | 13.33 (11.83-15.00) | <0.0001 | 13.45 (11.95-15.15) | <0.0001 |
| Treatment course intent | -- | -- | -- | -- | -- | -- | -- | -- |
| Curative | Ref | -- | Ref | -- | Ref | -- | Ref | -- |
| Oligomet. ablation | 4.18 (3.96-4.42) | <0.0001 | 1.83 (1.69-1.98) | <0.0001 | 1.83 (1.69-1.98) | <0.0001 | 1.82 (1.69-1.97) | <0.0001 |
| Palliative | 8.77 (8.45-9.10) | <0.0001 | 3.44 (3.25-3.64) | <0.0001 | 3.43 (3.24-3.63) | <0.0001 | 3.44 (3.25-3.65) | <0.0001 |
| Year treated | -- | -- | -- | -- | -- | -- | -- | -- |
| 2008-2015 | Ref | -- | Ref | -- | Ref | -- | Ref | -- |
| 2016-2023 | 0.74 (0.71-0.76) | <0.0001 | 0.81 (0.78-0.85) | <0.0001 | 0.81 (0.78-0.85) | <0.0001 | 0.82 (0.78-0.85) | <0.0001 |
| Radiation technique[e] | -- | -- | -- | -- | -- | -- | -- | -- |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 3D | Ref | -- | Ref | -- | Ref | -- | Ref | -- |
| IMRT | 0.69 (0.66-0.72) | <0.0001 | 0.75 (0.71-0.80) | <0.0001 | 0.75 (0.71-0.80) | <0.0001 | 0.75 (0.71-0.79) | <0.0001 |
| SBRT | 1.08 (1.00-1.16) | 0.061 | 0.55 (0.50-0.60) | <0.0001 | 0.55 (0.50-0.60) | <0.0001 | 0.55 (0.50-0.60) | <0.0001 |
| SRS | 2.58 (2.43-2.73) | <0.0001 | 0.71 (0.65-0.78) | <0.0001 | 0.71 (0.65-0.77) | <0.0001 | 0.71 (0.66-0.78) | <0.0001 |
| Brachytherapy | 0.20 (0.17-0.24) | <0.0001 | 0.31 (0.25-0.38) | <0.0001 | 0.31 (0.25-0.38) | <0.0001 | 0.32 (0.26-0.39) | <0.0001 |
| Electrons | 0.56 (0.49-0.63) | <0.0001 | 0.61 (0.53-0.71) | <0.0001 | 0.61 (0.53-0.71) | <0.0001 | 0.61 (0.53-0.71) | <0.0001 |
| Adaptive | 0.72 (0.60-0.86) | 0.00029 | 0.55 (0.45-0.66) | <0.0001 | 0.55 (0.45-0.66) | <0.0001 | 0.54 (0.45-0.66) | <0.0001 |

Two-sided Wald test P-values. Abbreviations: HR, hazard ratio; CI, confidence interval; Ref, reference; DCIS, ductal carcinoma in situ; oligomet., oligometastasis; 3D, 3-dimensional conformal; IMRT, intensity modulated radiotherapy; SBRT, stereotactic radiotherapy; SRS, stereotactic radiosurgery.

[a]n=38,183 with 13,769 events (63.9% censored)

[b]n=34,775 with 12,801 events (63.2% censored)

[c]n=37,200

[d]Excluded from multivariate analysis given significant confounding with cancer site variable

[e]n=35,706

**Extended Data Table 5. Diagnosis grouping according to ICD-9 or ICD-10 codes**

| ICD diagnosis | Diagnosis | Cancer site | ICD-9 codes | ICD-10 codes |
|---|---|---|---|---|
| Benign brain | Benign | Benign | 225.1, 225.2, 225.9, 225, 192.1, 747.81 | D32.9, D36.10, D36.11, D32.0, D33.3, Q28.2, D42.0, D35.2, D44.4 |
| Benign cardiovascular | Benign | Benign | 447.1 | I25.110, I73.9, I70.1, I70.212, I70.292, I70.90, I70.8, I47.2, I77.0, I74.4 |
| Benign HO/OA | Benign | Benign | 728.13, 213.6, 213.9, 215.9, 213.7, 215.3, 715.15, 728.1 | M89.8X9, M61.00, M61.49, M61.9, M61.50, M61.551, D16.11, M61.452, M16.12, M61.49, S73.004S, M25.8, M89.8X0, M61.552 |
| Benign keloid | Benign | Benign | 701.4 | L73.0, L91.0 |
| Benign other | Benign | Benign | 705.83 | D21.12, N62, M72.0, D21.21, D11.0, L73.2 |
| Bone met. | Bone met. | Bone met. | 198.5 | C79.51, C79.52 |
| Brain met. | Brain met. | Brain met. | 198.3 | C79.31, C79.32 |
| Other met. | Other met. | Other met. | 198.8, 196.9, 196.3, 198.2, 198.6, 198.7, 198.89, 196.2, 198.81, 198.1, 198.4, 196.6, 196.5, 198.82, 196.8 | C77.4, C77.3, C77.9, C77.5, C79.89, C79.40, C79.82, C78.4, C77.2, G95.20, C79.9, C79.49, C77.0, C77, C79.81, C79.2, C78.89, C79.72, C79.70, C78.6, C79.11, C76.3 |
| DCIS/LCIS[a] | DCIS | Breast | 233 | D05.12, D05.11, D05.01, D05.02, D05.81, D05.10, D05.80, D05.82, D05.90, D05.91, D05.92 |
| GI other prim.[a] | Other GI | GI | 153.3, 153.9, 153.6 | C18.4, C18.7, C76.2, C18.9, C78.5, C18.2, C18.1, C18.8 |
| GU other prim.[a] | Non-prostate GU | Other prim. | 186.9, 187.4, 187.2, 187.3 | C68.0, C62.12, C62.11, C60.9, C60.2 |
| GYN other prim.[a] | GYN | GYN | NA | C56.9, C56.1, C57.4, C57.00, C56.2 |
| Heme other[a] | Other Heme | Heme | 284.9, 238.75, 238.73, 284.8, 289.83, 238.79, 238.76, 284.09 | D61.9, D46.9, D75.81, D47.1, C94.6, D61.89, C96.A, D61.3, C88.0 |
| Hodgkin lymphoma[a] | Hodgkin lymphoma | Heme | 201.02, 201.41, 201.42, 201.44, 201.48, 201.4, 201.51, 201.52, 201.53, 201.55, 201.58, 201.5, 201.62, 201.64, 201.68, 201.6, 201.91, 201.92, 201.98, 201.9 | C81.01, C81.02, C81.04, C81.05, C81.08, C81.10, C81.11, C81.12, C81.13, C81.14, C81.18, C83.11, C83.12, C83.13, C81.14, C83.18, C81.20, C81.21, C81.24, C81.40, C81.70, C81.71, C81.72, C81.78, C81.79, C81.90, C81.91, C81.92, C81.94, C81.95, C81.96, C81.98, C81.99, C81.00 |
| HPV- OP SCC[a/b] | Other HN | HN | 146.1, 146.2, 146.4, 146.5, 146.7, 146.6, 146.8, 146.9, 146 | C10.2, C10.3, C10.9, C10.8, C09.0, C09.1, C09.9, C09.8, C01 |

| | | | | |
|---|---|---|---|---|
| Leukemia | Leukemia | Heme | 183, 204, 204.01, 204.02, 204.1, 204.21, 204.8, 205, 205.01, 205.02, 205.1, 205.11, 205.12, 205.21, 205.9, 206.01, 208.01, 208.02, 207.82 | C91.00, C91.01, C91.02, C92.00, C92.01, C92.02, C92.50, C92.51, C92.62, C92.A0, C92.A1, C92.A2, C93.00, C95.00, C95.01, C95.02, C91.51, C92.0, C91.0, C95.92, C92.90, C95.91 |
| Liver met. | Other met. | Other met. | 197.7 | C78.7 |
| Lung/thoracic met. | Lung met. | Other met. | 197.1, 197.2, 196.1, 197 | C78.00, C78.01, C78.02, C77.1, C78.2, C78.1, C78.39 |
| Male prim. breast[a] | Other prim. | Other prim. | 175, 175.90 | C50.929, C50.121, C50.921, C50.122, C50.922 |
| Melanoma[a] | Skin | Skin | 172.9, 172.3, 172.4, 172.5, 172.6, 172.7, 172.8, 172.2 | C43.11, C43.21, C43.30, C43.31, C43.39, C43.51, C43.59, C43.60, C43.61, C43.62, C43.72, C43.8, C43.9, C43.4 |
| Merkel cell[a] | Skin | Skin | 209.31, 209.32, 209.33, 209.34, 209.36 | C4A.10, C4A.11, C4A.112, C4A.22, C4A.30, C4A.31, C4A.39, C4A.4, C4A.51, C4A.52, C4A.59, C4A.61, C4A.60, C4A.62, C4A.70, C4A.71, C4A.72, C4A.8, C4A.9 |
| Merkel cell met. | Other met. | Other met. | 209.75 | C7B.1 |
| Multiple myeloma[a] | Multiple myeloma | Heme | 203, 203.01, 203.02, 238.6 | C90.00, C90.01, C90.02, C90.0, C90.30, C90.20 |
| NHL (aggressive) | NHL (aggressive) | Heme | 200.41, 200.48, 200.71, 200.78, 200.72, 200.75, 200.73, 202.7, 200.7, 200.4, 200.28, 200.68 | C83.10, C83.11, C83.16, C83.18, C83.19, C83.30, C83.31, C83.32, C83.33, C83.34, C83.35, C83.38, C83.39, C84.45, C84.48, C84.49, C84.61, C84.65, C84.69, C84.79, C84.09, C84.A9, C84.A, C93.10, C84.A5 |
| NHL (unspecified)[c] | NHL (aggressive or indolent) | Heme | 202.8, 202.88, 202.81, 202.85, 202.83 | C83.52, C83.70, C85.19, C85.80, C85.81, C85.83, C85.84, C85.85, C85.88, C85.89, C85.90, C85.9, C85.91, C85.92, C85.93, C85.94, C85.95, C85.96, C85.98, C85.99 |
| NHL (indolent) | NHL (indolent) | Heme | 200.31, 200.33, 200.34, 202.05 | C82, C82.00, C82.01, C82.02, C82.03, C82.04, C82.05, C82.06, C82.07, C82.08, C82.09, C82.10, C82.11, C82.12, C82.13, C82.14, C82.15, C82.16, C82.17, C82.18, C82.19, C82.20, C82.21, C82.22, C82.23, C82.24, C82.25, C82.26, C82.27, C82.28, C82.29, C82.30, C82.31, C82.32, C82.33, C82.34, C82.35, C82.36, C82.37, C82.38, C82.39, C82.40, C82.41, C82.42, C82.43, C82.44, C82.45, C82.46, C82.47, C82.48, C82.49, C82.50, C82.51, C82.52, C82.53, C82.54, C82.55, C82.56, C82.57, C82.58, C82.59, C82.60, C82.61, C82.62, C82.63, C82.64, C82.65, C82.66, C82.67, C82.68, C82.69, C82.70, C82.71, C82.72, C82.73, C82.74, C82.75, C82.76, C82.77, C82.78, C82.79, C82.80, C82.81, C82.82, C82.83, C82.84, C82.85, C82.86, C82.87, C82.88, C82.89, C82.90, C82.91, C82.92, C82.93, C82.98, C82.99, C83.00, C83.01, C83.09, C83.14, C83.80, C83.81, C83.82, C83.89, C83.99, C84.00, C84.01, C84.10, C84.A0, C85.10, C88.4, C91.10, C91.12 |

| | | | | |
|---|---|---|---|---|
| Other HN[a] | Other HN | HN | 141, 141.1, 141.2, 141.3, 141.4, 141.5, 141.6, 141.8, 141.9, 142, 142.1, 142.8, 142.9, 143.1, 144, 144.1, 144.8, 144.9, 145, 145.2, 145.3, 145.5, 145.6, 145.8, 145.9, 147.1, 147.2, 147.8, 147.9, 148.1, 148.3, 148.8., 148.9, 1489, 149.8, 160, 160.1, 160.2, 160.3, 160.4, 160.8, 160.9, 161, 161.1, 161.2, 161.3, 161.8, 161.9, 193, 149.9, 148.8, 210.2, 199.1, 140.1 | C00.0, C00.1, C00.4, C00.9, C02.0, C02.1, C02.2, C02.3, C02.4, C02.8, C02.9, C03.0, C03.1, C03.9, C04.0, C04.1, C04.8, C04.9, C05.0, C05.1, C05.2, C05.9, C06.0, C06.1, C06.2, C06.89, C06.9, C07, C08.0, C08.1, C08.9, C11.0, C11.1, C11.2, C11.8, C11.9, C12, C13.0, C13.2, C13.8, C13.9, C14.0, C14.8, C30.0, C31.0, C31.1, C31.8, C31.9, C32.0, C32.1, C32.8, C32.9, C73, C76.0, C85.819, C18.2, C32 |
| Other unspecified prim.[a] | Other prim. | Other prim. | 192.2, 194, 195, 170 | C7A.8, C7B.03, C47.9, C72.0, C26.1, C72.9, D48.9, C80.1, C70.1 |
| Prim. bladder/kidney[a] | Non-prostate GU | Other prim. | 188.1, 188.2, 188.3, 188.4, 188.5, 188.6, 188.8, 188.9, 189.1, 189.2, 189.3, 188, 189 | C67.0, C67.1, C67.2, C67.3, C67.4, C67.5, C64.1, C64.2, C64.9, C67.8, C67.9, C67.6, C74.02, C74.91, C68.9, C79.01, C79.71, C74.90, C65.1, C74.92, C74.01, C66.2, C74.00 |
| Prim. breast[a] | Breast | Breast | 174.1, 174.2, 174.3, 174.4, 174.5, 174.6, 174.8, 174.9, 174 | C50.912, C50.911, C50.411, C50.412, C50.812, C50.811, C50.111, C50.112, C50.212, C50.919, C50.511, C50.011, C50.211, C50.311, C50.611, C50.012, C50.312, C50.512, C50.612, C50.512, C50.612, C50.819 |
| Prim. esophageal/gastric[a] | Esophageal/Gastric | GI | 151.9, 151.3, 151.4, 151.5, 151.8, 151.1, 151.6, 150.9, 150.8, 150.5, 150.4, 150.3, 150.2, 150.1, 152, 150, 151 | C15.9, C15.5, C15.3, C15.4, C15.8, C16.2, C16.0, C16.1, C16.3, C16.4, C16.8, C16.9, C17.0 |
| Prim. GYN (cervix/vulva)[a] | GYN | GYN | 180.1, 180.2, 180.9, 180.8, 184.4, 184, 180 | C53.0, C53.1, C53.8, C53.9, C51.8, C51.9, C51.0, C52 |
| Prim. GYN uterine | GYN | GYN | 182.8, 179, 182 | C54.1, C54.8, C54.9, C54.2, C54.3, C55 |
| Prim. liver/GB[a] | Other GI | GI | 156.9, 156.8, 156.1, 155.1, 155, 156 | C22.0, C22.1, C22.2, C22.8, C22.9, C24.0, C23, C24.8, C24.9, C22 |
| Prim. lung/thorax[a] | Lung | Lung | 162.2, 162.3, 162.4, 162.5, 162.8, 162.9, 163.8, 163.9, 164.9, 162, 164, 163, 165.8, 164.8 | C34.11, C34.12, C34.31, C34.32, C34.90, C34.91, C34.92, C34.01, C34.02, C34.10, C34.80, C34.81, C34.82, C38.1, C38.3, C38.4, C45.0, C76.1, C34.2, C34.30, C33, C34.00, C7A.090, C37, C34.1, C34.9, C45.7, R91.1 |
| Prim. pancreas[a] | Pancreas | GI | 157.1, 157.2, 157.8, 157.9, 156.2, 157 | C25.1, C25.2, C25.3, C25.4, C25.7, C25.8, C25.9, C24.1, C25.0, C25 |
| Prim. prostate[a] | Prostate | Prostate | 185 | C61 |
| Prim. rectum/anus[a] | Rectal/Anal | GI | 154.1, 154.2, 154.3, 154.8, 154 | C20, C19, C21.0, C21.1, C21.8, C21 |
| Primary HGG/GBM | Brain (malignant) | Brain (malignant) | 191.1, 191.2, 191.3, 191.4, 191.5, 191.6, 191.7, 191.8, 191.9, 191 | C71.0, C71.1, C71.2, C71.3, C71.4, C71.5, C71.6, C71.7, C71.8, C71.9, C70.0, C70.9 |

| | | | | |
|---|---|---|---|---|
| Sarcoma[a] | Sarcoma | Sarcoma | 170.1, 170.2, 170.4, 170.6, 170.7, 170.9, 171, 171.2, 171.3, 171.4, 171.5, 171.6, 171.7, 171.8, 171.9 | C40.01, C40.20, C40.22, C40.21, C40.30, C40.90, C41.0, C41.1, C41.2, C41.3, C41.4, C41.9, C49.1, C49.10, C49.12, C49.2, C49.20, C49.21, C49.22, C49.3, C49.4, C49.5, C49.6, C49.8, C49.80, C49.81, C49.82, C76.52, C49.0, C48.0, C49.9, C40.32, C76.51, C49.11, C76.41, C48.2, C46.0 |
| SCC/BCC[a] | Skin | Skin | 173.32, 173.31, 173.3, 173.01, 173.02, 173.09, 173.1, 173.12, 173.2, 173.21, 173.22, 173.4, 173.41, 173.42, 173.49, 173.5, 173.52, 173.6, 173.61, 173.62, 173.7, 173.71, 173.72, 173.8, 173.82, 173.89, 173.9, 173.92, 173.99, 173.39 | C44.00, C44.01, C44.02, C44.111, C44.112, C44.121, C44.202, C44.209, C44.212, C44.219, C44.221, C44.222, C44.229, C44.292, C44.299, C44.300, C44.301, C44.309, C44.310, C44.311, C44.319, C44.320, C44.321, C44.329, C44.40, C44.41, C44.42, C44.509, C44.519, C44.520, C44.529, C44.590, C44.599, C44.602, C44.612, C44.619, C44.621, C44.622, C44.629, C44.719, C44.721, C44.722, C44.729, C44.80, C44.81, C44.89, C44.90, C44.91, C44.92, C44.99, C44.390, C44.391, C44.399 |

Abbreviations: ICD, international classification of diseases; HO/OA, heterotopic ossification/osteoarthritis; met., metastasis; DCIS, ductal carcinoma in situ; LCIS, lobular carcinoma in situ; GI, gastrointestinal; prim., primary (non-metastatic); GU, genitourinary; GYN, gynecological; NA, not applicable; Heme, hematological; HPV, human papillomavirus; OP, oropharynx; SCC, squamous cell carcinoma; NHL, non-Hodgkin lymphoma; HN, head and neck; GB, gall bladder; HGG, high grade glioma; GBM, glioblastoma; BCC, basal cell carcinoma

[a]Classified as metastasis if diagnosis description specifies “metastatic,” “metastasis,” “metastases,” “mets,” “met,” or “stage IV”

[b]Classified as HPV+ OP if diagnosis description specifies “p16+,” “p16 positive,” “HPV positive,” “HPV+,” ”+HPV,” “HPV associated,” “HPV related,” or “HPV mediated”

[c]Classified as NHL (aggressive) if diagnosis description specifies “DLBCL,” “large cell,” “diffuse large,” “anaplastic,” “T cell,” “mantle,” “high grade,” “G3,” or “grade 3,” versus NHL (indolent) if diagnosis description specified “follicular,” “marginal zone,” “indolent,” “low grade,” “G1,” or “grade 1”

TRIPOD+AI

Version: 11-January-2024

| Section/Topic | Item | Development / evaluation[1] | Checklist item | Reported on page |
|---|---|---|---|---|
| **TITLE** | | | | |
| *Title* | 1 | D;E | Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted | 1 |
| **ABSTRACT** | | | | |
| *Abstract* | 2 | D;E | See TRIPOD+AI for Abstracts checklist | 2-3 |
| **INTRODUCTION** | | | | |
| *Background* | 3a | D;E | Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models | 3-4 |
| | 3b | D;E | Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users (e.g., healthcare professionals, patients, public) | 3-4 |
| | 3c | D;E | Describe any known health inequalities between sociodemographic groups | 3-4 |
| *Objectives* | 4 | D;E | Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both) | 3-4 |
| **METHODS** | | | | |
| *Data* | 5a | D;E | Describe the sources of data separately for the development and evaluation datasets (e.g., randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data | 4-6 |
| | 5b | D;E | Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up | 4-6 |
| *Participants* | 6a | D;E | Specify key elements of the study setting (e.g., primary care, secondary care, general population) including the number and location of centres | 4-6 |
| | 6b | D;E | Describe the eligibility criteria for study participants | 4-6 |
| | 6c | D;E | Give details of any treatments received, and how they were handled during model development or evaluation, if relevant | 4-6 |
| *Data preparation* | 7 | D;E | Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups | 4-6 |
| *Outcome* | 8a | D;E | Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups | 4-6 |
| | 8b | D;E | If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors | 4-6 |
| | 8c | D;E | Report any actions to blind assessment of the outcome to be predicted | 4-6 |
| *Predictors* | 9a | D | Describe the choice of initial predictors (e.g., literature, previous models, all available predictors) and any pre-selection of predictors before model building | 4-6 |
| | 9b | D;E | Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors) | 4-6 |
| | 9c | D;E | If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors | 4-6 |
| *Sample size* | 10 | D;E | Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation | 4-6 |
| *Missing data* | 11 | D;E | Describe how missing data were handled. Provide reasons for omitting any data | 4-6 |
| *Analytical methods* | 12a | D | Describe how the data were used (e.g., for development and evaluation of model performance) in the analysis, including whether the data were partitioned, considering any sample size requirements | 4-6 |
| | 12b | D | Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation). | 4-6 |
| | 12c | D | Specify the type of model, rationale[2], all model-building steps, including any hyperparameter tuning, and method for internal validation | 4-6 |
| | 12d | D;E | Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters (e.g., hospitals, countries). See TRIPOD-Cluster for additional considerations[3] | 4-6 |
| | 12e | D;E | Specify all measures and plots used (and their rationale) to evaluate model performance (e.g., discrimination, calibration, clinical utility) and, if relevant, to compare multiple models | 4-6 |
| | 12f | E | Describe any model updating (e.g., recalibration) arising from the model evaluation, either overall or for particular sociodemographic groups or settings | 4-6 |
| | 12g | E | For model evaluation, describe how the model predictions were calculated (e.g., formula, code, object, application programming interface) | 4-6 |
| *Class imbalance* | 13 | D;E | If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions | 4-6 |
| *Fairness* | 14 | D;E | Describe any approaches that were used to address model fairness and their rationale | 4-6 |
| *Model output* | 15 | D | Specify the output of the prediction model (e.g., probabilities, classification). Provide details and rationale for any classification and how the thresholds were identified | 4-6 |

[1] D=items relevant only to the development of a prediction model; E=items relating solely to the evaluation of a prediction model; D;E=items applicable to both the development and evaluation of a prediction model

[2] Separately for all model building approaches.

[3] TRIPOD-Cluster is a checklist of reporting recommendations for studies developing or validating models that explicitly account for clustering or explore heterogeneity in model performance (eg, at different hospitals or centres). Debray et al, BMJ 2023; 380: e071018 [DOI: 10.1136/bmj-2022-071018]

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|---|---|---|---|---|
| *Training versus evaluation* | 16 | D;E | Identify any differences between the development and evaluation data in healthcare setting, eligibility criteria, outcome, and predictors | 4-6 |
| *Ethical approval* | 17 | D;E | Name the institutional research board or ethics committee that approved the study and describe the participant-informed consent or the ethics committee waiver of informed consent | 4-6 |
| **OPEN SCIENCE** | | | | |
| *Funding* | 18a | D;E | Give the source of funding and the role of the funders for the present study | 6 |
| *Conflicts of interest* | 18b | D;E | Declare any conflicts of interest and financial disclosures for all authors | 12 |
| *Protocol* | 18c | D;E | Indicate where the study protocol can be accessed or state that a protocol was not prepared | NA |
| *Registration* | 18d | D;E | Provide registration information for the study, including register name and registration number, or state that the study was not registered | NA |
| *Data sharing* | 18e | D;E | Provide details of the availability of the study data | 12-13 |
| *Code sharing* | 18f | D;E | Provide details of the availability of the analytical code[4] | 13 |
| **PATIENT & PUBLIC INVOLVEMENT** | | | | |
| *Patient & Public Involvement* | 19 | D;E | Provide details of any patient and public involvement during the design, conduct, reporting, interpretation, or dissemination of the study or state no involvement. | NA |
| **RESULTS** | | | | |
| *Participants* | 20a | D;E | Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time. A diagram may be helpful. | 6-10 |
| | 20b | D;E | Report the characteristics overall and, where applicable, for each data source or setting, including the key dates, key predictors (including demographics), treatments received, sample size, number of outcome events, follow-up time, and amount of missing data. A table may be helpful. Report any differences across key demographic groups. | 6-10 |
| | 20c | E | For model evaluation, show a comparison with the development data of the distribution of important predictors (demographics, predictors, and outcome). | 6-10 |
| *Model development* | 21 | D;E | Specify the number of participants and outcome events in each analysis (e.g., for model development, hyperparameter tuning, model evaluation) | 6-10 |
| *Model specification* | 22 | D | Provide details of the full prediction model (e.g., formula, code, object, application programming interface) to allow predictions in new individuals and to enable third-party evaluation and implementation, including any restrictions to access or re-use (e.g., freely available, proprietary)[5] | 4-6 |
| *Model performance* | 23a | D;E | Report model performance estimates with confidence intervals, including for any key subgroups (e.g., sociodemographic). Consider plots to aid presentation. | 6-10 |
| | 23b | D;E | If examined, report results of any heterogeneity in model performance across clusters. See TRIPOD Cluster for additional details[3]. | 6-10 |
| *Model updating* | 24 | E | Report the results from any model updating, including the updated model and subsequent performance | NA |
| **DISCUSSION** | | | | |
| *Interpretation* | 25 | D;E | Give an overall interpretation of the main results, including issues of fairness in the context of the objectives and previous studies | 10-11 |
| *Limitations* | 26 | D;E | Discuss any limitations of the study (such as a non-representative sample, sample size, overfitting, missing data) and their effects on any biases, statistical uncertainty, and generalizability | 10-11 |
| *Usability of the model in the context of current care* | 27a | D | Describe how poor quality or unavailable input data (e.g., predictor values) should be assessed and handled when implementing the prediction model | 10-11 |
| | 27b | D | Specify whether users will be required to interact in the handling of the input data or use of the model, and what level of expertise is required of users | 10-11 |
| | 27c | D;E | Discuss any next steps for future research, with a specific view to applicability and generalizability of the model | 10-11 |

From: Collins GS, Moons KGM, Dhiman P, et al. *BMJ* 2024;385:e078378. doi:10.1136/bmj-2023-078378

[4] This relates to the analysis code, for example, any data cleaning, feature engineering, model building, evaluation.
[5] This relates to the code to implement the model to get estimates of risk for a new individual.